

Geometric Observables for Financial Regime Detection: A Spectral Metric Learning Approach

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Abstract

We extract geometric observables from a spectral metric learning framework—Quantum Cognition Machine Learning (QCML; the name is historical—the mathematics is spectral geometry, not quantum physics)—and evaluate their ability to detect financial regime transitions. QCML encodes data as quasi-coherent states in a Hilbert space via an error Hamiltonian; the induced Fubini–Study metric and Berry curvature yield three unsupervised scalar observables: the Berry curvature increment, the quantum Fisher information (QFI) determinant, and a multi-lag fidelity measure.

In a study of 14 historical crises (2000–2024) with per-crisis causal preprocessing (scaler, PCA, and operators fitted only on pre-crisis data), all three geometric observables are retained in the 90% Model Confidence Set (Hansen et al., 2011), placing them in the set of methods statistically indistinguishable from the best—alongside CUSUM, the supervised Random Forest (RF) baseline, and seven other detectors; only BOCPD is eliminated. Bootstrap rank confidence intervals show Berry curvature increment (rank 4.79, CI [3.21, 6.43]) overlapping CUSUM (rank 4.57) and RF (rank 5.64, CI [4.07, 7.14]). An oracle selector achieves median Cohen’s $d = 1.15$, a 68% improvement over the best single method (CUSUM, $d = 0.69$), with complementarity score 0.73, confirming that geometric and classical methods capture non-overlapping crisis information.

Berry curvature increment achieves the highest geometric median Cohen’s d (0.63), outperforming RF (0.37) and trailing only CUSUM (0.69). The best geometric method exceeds RF on 10 of 14 crises—precisely those where causal training labels are sparse or absent. An equal-weight causal fusion of all three geometric observables achieves median $d = 0.66$ across 12 crises, improving on the best individual detector (Berry, $d = 0.58$) by 14% and on CUSUM ($d = 0.63$) by 6%, without requiring any crisis labels.

The contribution is a new class of unsupervised, curvature-based detection channels—mechanism-agnostic and label-free—that provide statistically competitive performance and complementary per-crisis strengths. In walk-forward validation with FAR-calibrated thresholds, Multi-Lag Fidelity detects the majority of crises (3–6/7 across validation runs) with a median delay of 16–23 trading days and 30% fewer false alarms than RF (2.5 vs. 3.6/yr). An online detection pipeline with corrected signal polarity achieves AUC-ROC = 0.84, resolving an initial directional mismatch (crisis manifold contraction rather than expansion) that had degraded the naïve pipeline to chance. Four formal results (smoothness, Chern quantization, curvature–gap bound, QFI–metric identity) are empirically validated on the full dataset, with the curvature–gap bound holding at 100% of sampled points and the QFI–metric identity verified to machine precision. All effect sizes include bootstrap 95% CIs ($n = 5,000$).

Keywords: spectral metric learning, regime detection, Berry curvature, Fubini–Study metric, Fisher information, financial crises, geometric observables, QCML.

1 Introduction

Detecting regime transitions—shifts between qualitatively different market states—is a central problem in quantitative finance. Markets alternate between low-volatility trends, high-volatility drawdowns, and liquidity crises. Classical approaches include Hidden Markov Models [3], Bayesian Online Changepoint Detection [32], and supervised classifiers. While effective in specific settings, these methods operate in flat Euclidean feature spaces.

Quantum Cognition Machine Learning (QCML) [1]—introduced in industry white papers [35, 36] and subsequently validated in peer-reviewed work [1]—encodes data points as unit vectors in a finite-dimensional Hilbert space and represents features as Hermitian operators. The map $x \mapsto |\psi(x)\rangle$ from feature space to projective Hilbert space $\mathbb{CP}^{n-1}$ endows the data manifold with a Fubini–Study metric, Berry curvature, spectral gap, and quantum Fisher information. Candelori et al. [1] showed that this geometry robustly captures intrinsic dimension; Abanov et al. [28] developed the mathematical theory. Financial applications include forecasting [36], similarity learning [29, 30], and biomedical prediction [31].

Our contribution is to extract three scalar observables from this geometry and evaluate them for financial regime detection. The evaluation is framed as *crisis-window separability*: given a score time series, we ask whether the distribution of scores during a known crisis window is statistically separated from the non-crisis distribution. This is an *offline event study*—it measures sensitivity to regime transitions, not real-time predictive power. A supplementary walk-forward benchmark (Section 5.4) tests causal detection with strictly past-fitted preprocessing.

Clarification on “unsupervised.” Score construction is unsupervised: the QCML observables require no labeled crisis data to compute z-scores (Algorithm 1). Hyperparameters (Hilbert dimension, PCA components, operator method, rolling window) are selected via Optuna TPE with a consistency penalty (Berry: $n = 6$, $p = 8$, $w = 15$; QFI: $n = 8$, $p = 15$, $w = 20$; Multi-Lag: $n = 4$, $p = 8$, $w = 20$); operator method is selected by ablation on historical data (Section 5.9). We use “unsupervised score construction with HPO-optimized hyperparameters” throughout.

Contributions.

1. Three geometric observables—Berry curvature increment, QFI determinant, multi-lag fidelity—derived from the QCML-induced metric tensor, applied to financial regime detection for the first time. All three are retained in the 90% Model Confidence Set, placing them in the set of methods statistically indistinguishable from the best (Table 10).
2. An equal-weight causal fusion of all three geometric observables (Section 5.8) that achieves median $d = 0.66$ across 12 crises, improving on the best individual detector by 14% and on CUSUM by 6%. The fusion requires no crisis labels and uses a parameter-free equal-weight average, demonstrating that geometric observables provide complementary detection channels whose combination improves upon any individual method.
3. Evidence that unsupervised geometric methods provide *complementary* detection with lower false alarm rates: in walk-forward validation, Multi-Lag Fidelity detects the majority of crises with 30% fewer false alarms than RF (2.5 vs. 3.6/yr; Table 8), trading detection speed for precision.
4. A comprehensive evaluation framework including per-crisis causal preprocessing, Model Confidence Set analysis, Bayesian pairwise tests, bootstrap rank CIs, detection metrics (F1,

AUC-PR, delay, FAR), and expanding-window walk-forward validation—providing a multi-dimensional assessment rather than reliance on a single summary statistic.

5. Empirical validation of four formal results (Theorems 1–4, Proposition 2) on 20 years of market data (Section 5.11). The curvature–gap bound holds at 100% of sampled points; the QFI–metric identity holds to machine precision ($\text{RMSE} = 1.84 \times 10^{-11}$).
6. To our knowledge, this is the first independent academic evaluation of QCML geometry for financial regime detection; prior QCML publications are either from the framework’s developers [1, 28] or industry white papers [35, 36].

We emphasize an honest framing of these results. CUSUM retains the highest median Cohen’s d (0.69), but Berry curvature (0.63) outperforms RF (0.37). Bayesian pairwise tests show $P(\text{Berry} > \text{RF}) = 0.48$, though posteriors are broad given only 14 crises. In walk-forward validation, RF detects faster (5-day vs. 16–23 day median delay) and more completely (7/7 vs. majority of crises). However, aggregate effect size and speed are only two dimensions of evaluation. Geometric methods are statistically indistinguishable from the best in the Model Confidence Set, produce fewer false alarms (2.5 vs. 3.6/yr) without requiring labels, and the best geometric observable wins 10 of 14 crises head-to-head against RF—precisely those where supervised training data is sparse or absent (e.g., 2018 Q4: Berry $d = 2.00$ vs. RF $d = 0.28$). The practical contribution is a new class of manifold-intrinsic, label-free detection channels that complement rather than replace supervised methods [10, 11].

1.1 Related Work

Several non-Euclidean frameworks have been applied to financial data. We briefly survey the most relevant and position our contribution.

Information geometry. Amari and Nagaoka [7] introduced the Fisher–Rao metric on statistical manifolds; Brody and Hughston [15] applied it to interest rate term structures, and Taylor [16] used Fisher distance for clustering return distributions. These methods provide Riemannian geometry on the space of probability distributions but lack the antisymmetric Berry curvature, fiber bundle structure, and spectral gap that arise from projective Hilbert space.

Optimal transport. Horvath et al. [17] cluster market regimes via Wasserstein distance. Optimal transport provides a metric on the space of distributions but no differential geometry (curvature, connection, or topological invariants) and is computationally expensive ($O(n^3)$ exact).

TDA and persistent homology. Gidea and Katz [18] detect crash precursors via L^p -norms of persistence landscapes. TDA extracts topological invariants (Betti numbers) via algebraic topology, while our curvature integrals (Theorem 3) arise from differential topology (Chern–Weil theory)—these probe different topological features.

Network curvature. Sandhu et al. [19] use Ollivier–Ricci curvature on stock correlation networks as a systemic risk indicator—the closest curvature-based regime detector to our work. Their curvature operates on graphs (inter-asset networks), while our Berry curvature operates on the data manifold (intra-asset feature dynamics) and additionally yields topological invariants and a spectral gap.

Random matrix theory. Laloux et al. [20] and Plerou et al. [21] use the Marchenko–Pastur distribution to separate signal from noise in correlation matrices. Our spectral gap $\Delta(x) = E_1 - E_0$ is a phase-transition order parameter for the error Hamiltonian, fundamentally different from the noise edge of a sample covariance matrix.

Rough path signatures and geometric deep learning. Issa and Horvath [33] detect regimes via signature-based MMD tests; signatures provide universal path features but lack geometric interpretability. Bronstein et al. [34] systematize gauge-equivariant neural architectures but impose geometric priors rather than discovering geometry from data.

Relationship to the QCML literature. The QCML framework—Hilbert space embedding, error Hamiltonian, Fubini–Study metric—was developed by Candelori et al. [1] and Abanov et al. [28]; the financial forecasting application was proposed in industry white papers [35, 36] (not peer-reviewed). Our contribution is not the framework itself but three specific applications: (i) extracting Berry curvature increment, QFI determinant, and multi-lag fidelity as standalone regime-detection observables; (ii) the first head-to-head comparison against established baselines (RF, HMM, CUSUM, BOCPD) with proper statistical testing (MCS, Bayesian pairwise tests, bootstrap rank CIs); and (iii) a comprehensive robustness framework including walk-forward validation and multi-asset generalization.

Table 1 summarizes these distinctions. Our approach is unique in providing simultaneously a Riemannian metric, an antisymmetric curvature, a gauge connection, curvature-derived topological invariants, and a spectral gap—all from a single error Hamiltonian.

Table 1: Non-Euclidean approaches to financial regime detection. Columns indicate which geometric tools each framework provides.

Framework	Metric	Curvature	Gauge	Topol. Invt.	Spect. Gap
Info. Geometry	Fisher–Rao	Sectional	α -conn.	—	—
Optimal Transport	Wasserstein	—	—	—	—
TDA	—	—	—	Betti	—
Network Ricci	—	Ollivier	—	—	—
Random Matrix Thy.	—	—	—	—	MP edge
Rough Path Sig.	Sig. kernel	—	—	—	—
QCML (ours)	Fubini–Study	Berry	U(1)	Chern	$\Delta(x)$

2 QCML Framework

We work in a finite-dimensional Hilbert space $\mathcal{H} \cong \mathbb{C}^n$. A *state* is a unit vector $|\psi\rangle \in \mathcal{H}$, and an *observable* is a Hermitian operator $A = A^\dagger$ [1, 28].

Given $x = (x_1, \dots, x_p) \in \mathbb{R}^p$ and Hermitian operators $\{A_k\}_{k=1}^p$, the QCML *error Hamiltonian* [1, 35] is:

$$H(x) = \frac{1}{2} \sum_{k=1}^p (A_k - x_k I)^2, \quad (1)$$

where I is the $n \times n$ identity. The *quasi-coherent state* at x is the ground state:

$$|\psi(x)\rangle = \arg \min_{|\phi\rangle, \|\phi\|=1} \langle \phi | H(x) | \phi \rangle. \quad (2)$$

The pullback of the Fubini–Study metric defines the *quantum metric tensor* [6]:

$$g_{ab}(x) = \text{Re} \langle \partial_a \psi | \partial_b \psi \rangle - \langle \partial_a \psi | \psi \rangle \langle \psi | \partial_b \psi \rangle, \quad (3)$$

where indices $a, b \in \{0, \dots, p-1\}$ run over PCA components, and the *Berry curvature* [2, 28]:

$$F_{ab}(x) = -2 \text{Im} \langle \partial_a \psi | \partial_b \psi \rangle. \quad (4)$$

Theorem 1 (Smoothness [1]). *If the spectral gap $\Delta(x) = E_1(x) - E_0(x) > 0$ on an open set $U \subset \mathbb{R}^p$, then $x \mapsto |\psi(x)\rangle$, $g_{ab}(x)$, and $F_{ab}(x)$ are C^∞ on U .*

Proof sketch. By the implicit function theorem applied to the eigenvalue equation $H(x)|\psi\rangle = E_0|\psi\rangle$, $\langle\psi|\psi\rangle = 1$; the non-degenerate spectral gap ensures invertibility of the relevant Jacobian. See Appendix A for the full argument. $\square$

Operator construction. We use *PCA-inspired* operators: Pauli-basis matrices scaled by PCA eigenvalues, $A_k = \sqrt{\lambda_k} \sigma_k$. An ablation comparing random, PCA-inspired, and learned-scaling operators appears in Section 5.9. Full QCML gradient-descent training [35, 36] is left for future work.

Proposition 2 (Fisher information interpretation). *For the pure states $|\psi(x)\rangle$ arising as ground states of $H(x)$, the quantum metric tensor (3) satisfies $4g_{ab}(x) = [F_Q]_{ab}(x)$, where F_Q is the quantum Fisher information matrix [13]. Consequently:*

- (i) *The QFI determinant observable (Eq. 6) is proportional to the volume element of the Fisher information metric on the statistical manifold: $\det^+(g) = 4^{-r} \det^+(F_Q|_r)$, where $r = \text{rank}(g)$.*
- (ii) *The quantum Cramér–Rao bound $\text{Var}(\hat{x}_a) \geq \frac{1}{4}[g^{-1}]_{aa}$ implies that the quantum metric controls the ultimate precision for estimating data coordinates from state measurements [14].*
- (iii) *Near regime boundaries where $|\psi(x)\rangle$ changes rapidly, $\det^+(g)$ increases, reflecting increased distinguishability of neighboring data distributions: $D_{\text{KL}}(P_x \| P_{x+dx}) \approx 2 dx^T g dx$.*

Proof. Part (i) follows from the Braunstein–Caves theorem [13] and Proposition 5. Part (ii) is the quantum Cramér–Rao bound [14]. Part (iii) follows from the local expansion of KL divergence in terms of the Fisher metric [7]. $\square$

3 Geometric Observables

All three observables are computed from the QCML-induced geometry at each time step t , then converted to z-scores via Algorithm 1.

3.1 Berry Curvature Increment

The rate of change of Berry curvature between consecutive time steps:

$$\dot{\gamma}(t) = |F_{01}(x_t) - F_{01}(x_{t-1})|, \quad (5)$$

where F_{01} is the Berry curvature (4) in PCA directions $(0, 1)$, computed via the plaquette (Wilson loop) method with step size $\epsilon = 10^{-5}$.

Hyperparameters. Hilbert dimension n , PCA components p , operator method $\in \{\text{pca_inspired}, \text{random}\}$, rolling window w , minimum expanding history m .

3.2 QFI Determinant

The pseudo-determinant of the quantum metric tensor:

$$\det^+(g(x_t)) = \prod_{\lambda_i > \epsilon_{\text{tol}}} \lambda_i, \quad (6)$$

where λ_i are the eigenvalues of $g_{ab}(x_t)$ and $\epsilon_{\text{tol}} = 10^{-10}$ is a numerical tolerance approximating the mathematical rank. Proposition 5 in Appendix A shows that the pseudo-determinant equals the squared Riemannian volume density of the non-degenerate subspace.

Hyperparameters. Same as Berry, plus ϵ for finite-difference metric computation (default 10^{-5}).

3.3 Multi-Lag Fidelity

Weighted average infidelity across time lags:

$$\bar{\mathcal{I}}(t) = \sum_j w_j (1 - |\langle \psi(x_t) | \psi(x_{t-\delta_j}) \rangle|^2), \quad (7)$$

with $\delta \in \{1, 3, 5, 10\}$ and weights $w = [0.4, 0.3, 0.2, 0.1]$. Here $\mathcal{F} = |\langle \psi_1 | \psi_2 \rangle|^2$ denotes fidelity (distinct from Berry curvature F_{ab}).

Hyperparameters. Same as Berry, plus lag set and weights.

3.4 Z-Score Thresholding (Algorithm 1)

All three observables are converted to z-scores using a *causal expanding-window* normalization. This is the “adaptive thresholding” referenced in Algorithm 3.

Algorithm 1 Causal Expanding-Window Z-Score

Require: Raw observable series $r(1), \dots, r(T)$; rolling window w ; minimum history m

Ensure: Z-score series $z(m), \dots, z(T)$

- 1: Smooth: $s(t) = \frac{1}{w} \sum_{i=0}^{w-1} r(t-i)$ for $t \geq w$
 - 2: **for** $t = m, \dots, T$ **do**
 - 3: $\mu(t) = \text{mean}(s(1), \dots, s(t-1))$ ▷ expanding mean of past only
 - 4: $\sigma(t) = \text{std}(s(1), \dots, s(t-1))$ ▷ expanding std of past only
 - 5: $z(t) = (s(t) - \mu(t)) / \sigma(t)$
 - 6: **end for**
-

Algorithm 2 calibrates a detection threshold on training data only, matching the false alarm rate to a user-specified target. We use $\alpha = 1$ alarm/year as the default. This avoids the scale mismatch of a fixed $z > 2.0$ threshold across methods whose score distributions differ.

Defaults: $w = 20$, $m = 60$; tuned via the HPO protocol in Section 4.2. No future data enters the z-score computation.

Algorithm 2 FAR-Calibrated Threshold

Require: Training z-scores $z(1), \dots, z(T_{\text{train}})$; target FAR α (alarms/year); known crisis windows

Ensure: Threshold τ

- 1: Remove crisis-window indices from z to obtain z_{normal}
 - 2: Binary search: find smallest τ such that $\text{FAR}(z_{\text{normal}} > \tau) \leq \alpha$
 - 3: **return** τ
-

4 Experimental Protocol

4.1 Data and Crisis Definitions

Daily OHLCV data for SPY and DIA (2005–2024) from Yahoo Finance via the `yfinance` library (v0.2.x). Features: log returns, 5/20-day rolling volatility, 5/20-day momentum, cross-correlation, and cross-dispersion (13 raw features), enriched to 52 features via rolling statistics (mean, std, min, max over a 20-day lookback), then reduced to $p = 15$ PCA components and L^2 -normalized.

We evaluate on 12 historical crises (Table 2). Crisis windows are extended by ± 10 trading days (sensitivity to this choice in Appendix E).

Table 2: Crisis periods. Categories defined *a priori* for the interaction test (Section 5.7).

Crisis	Period	Category	Rationale
2000 Dot-com Bust	2000-03 to 2000-12	Conventional	Valuation bubble
2001 September 11	2001-09 to 2001-10	Conventional	Exogenous shock
2007 Quant Meltdown	2007-08 to 2007-09	Conventional	Resembles LTCM
2008 GFC	2008-09 to 2009-03	Conventional	Credit crisis
2010 Flash Crash	2010-05 to 2010-06	Conventional	Resembles 1987
2011 Euro Crisis	2011-07 to 2011-10	Conventional	Sovereign debt
2015 China Crash	2015-07 to 2015-09	Conventional	EM contagion
2020 COVID	2020-02 to 2020-04	Conventional	Exogenous shock
2018 Volmageddon	2018-01 to 2018-04	Novel	Short-vol blowup
2018 Q4 Selloff	2018-10 to 2018-12	Novel	Algo-driven
2019 Repo Crisis	2019-09 to 2019-10	Novel	Plumbing crisis
2022 Rate Hikes	2022-01 to 2022-10	Novel	Fastest in 40yr
2023 SVB	2023-03 to 2023-04	Novel	Social-media bank run
2024 Carry Unwind	2024-07 to 2024-08	Novel	Yen carry

4.2 Methods and HPO Protocol

Table 3 documents all methods and their search spaces. For the main comparison (Table 4), geometric methods use HPO-optimized hyperparameters (Berry: $n = 6$, $p = 8$, $w = 15$, random operators; QFI: $n = 8$, $p = 15$, $w = 20$, PCA-inspired; Multi-Lag: $n = 4$, $p = 8$, $w = 20$, PCA-inspired) selected by Optuna TPE with consistency penalty (see below). Classical baselines use their standard defaults. RF uses leave-one-crisis-out labels.

Random Forest protocol. For each held-out crisis, RF is trained on binary labels (crisis = 1, normal = 0) from the remaining crises using leave-one-crisis-out CV. Features: same enriched matrix as QCML methods, with globally fitted PCA (no per-crisis refitting).

Table 3: Hyperparameter search spaces. Main results use fixed defaults (bold); ablation explores alternatives. Enriched features (4d rolling stats) for all methods.

Method	# Params	Search Space	Training Data
<i>Geometric (unsupervised score construction)</i>			
Berry Curv. Incr.	4	$n \in \{4, 8, 16\}$, $p \in \{10, 15, 20\}$, $op \in \{\text{rnd}, \text{pca}\}$, $w \in \{10, 20, 30\}$	Full (or expanding)
QFI Determinant	4	same as Berry	Full (or expanding)
Multi-Lag Fidelity	4	same as Berry	Full (or expanding)
QCML Chern	3	$n \in \{4, 8\}$, $p \in \{5, 10, 15\}$, $window \in \{10, 20\}$	Full
Geo. Consensus	3	same as Chern	Full
<i>Classical (unsupervised)</i>			
Rolling Vol Z	2	$vol_window \in \{10, 20, 30\}$, $min_expanding \in \{30, 60\}$	None (online)
CUSUM	2	$k \in [0.3, 1.0]$, $burn_in \in \{30, 60, 90\}$	Burn-in period
HMM (2-state)	2	$cov \in \{\text{full}, \text{diag}\}$, $n_iter \in \{50, 100, 200\}$	Full
BOCPD	1	$hazard \in [50, 500]$	None (online)
Isolation Forest	2	$n_est \in [50, 300]$, $contam \in [0.01, 0.15]$	Full
<i>Supervised</i>			
Random Forest	2	$n_est \in [50, 300]$, $depth \in [5, 20]$	LOCO labels

Per-crisis causal preprocessing. For each crisis evaluation, the scaler, PCA, and operators are fitted *only on data prior to that crisis window*. Specifically, the causal cutoff is set at crisis start minus 10 calendar days; all preprocessing (standardization, PCA projection, operator fitting) uses only rows $1, \dots, t_{\text{cutoff}}$. Scores are then computed on the full timeline. The 20-day enrichment lookback provides an additional implicit buffer of approximately 19 trading days, yielding an effective cutoff of ~ 33 trading days before crisis onset. Classical baselines are similarly trained only on pre-crisis data. RF uses leave-one-crisis-out labels restricted to the causal window. This eliminates lookahead in preprocessing.

Statistical evaluation. Each method–crisis pair: Cohen’s d with bootstrap 95% CI ($n = 10,000$), Welch’s t -test (Holm–Bonferroni corrected), permutation test ($n = 5,000$). Multi-method: Friedman rank test.

HPO-optimized hyperparameters. The main results (Table 4) use hyperparameters selected via Optuna TPE (100 trials) with a train/validation split (pre-2020/post-2020) and a consistency penalty $mean_d - 0.3 \cdot std_d$ that rewards stable detection across diverse crisis types. The search space covers $n \in \{4, 6, 8\}$, $p \in \{5, 8, 10, 15\}$, $w \in \{10, 15, 20\}$ with operator method fixed by prior ablation (Section 5.9). The regularized HPO reduces the train–validation gap to at most 0.20 (Berry: $0.68 \rightarrow 0.49$; MLF: $0.53 \rightarrow 0.33$), with QFI exhibiting *negative* overfitting (val $d = 0.60$ exceeds train $d = 0.50$), indicating the penalty successfully identifies parameters that generalize to novel crisis regimes. Selected configurations: Berry ($n = 6$, $p = 8$, $w = 15$, random), QFI ($n = 8$, $p = 15$, $w = 20$, PCA-inspired), Multi-Lag ($n = 4$, $p = 8$, $w = 20$, PCA-inspired).

4.3 Detection Pipeline

Computational cost. Berry curvature increment: 0.77s; Multi-Lag Fidelity: 0.26s; QFI Determinant: 5.95s; Random Forest: 1.07s ($T = 3,447$, single CPU, $n = 8$, $p = 15$).

Algorithm 3 QCML Geometric Regime Detection

Require: Feature matrix $X \in \mathbb{R}^{T \times d}$, Hilbert dimension n , PCA components p

Ensure: Z-scored observable time series

- 1: Standardize X ; PCA to p components; L^2 -normalize
 - 2: Fit operators $\{A_k\}$ via PCA-inspired method (or expanding-window refit)
 - 3: **for** $t = 1, \dots, T$ **do**
 - 4: Solve $|\psi(x_t)\rangle =$ ground state of $H(x_t)$ (1)
 - 5: Compute raw observables: $\dot{\gamma}(t)$ (5), $\det^+(g)$ (6), $\bar{I}(t)$ (7)
 - 6: **end for**
 - 7: Convert to z-scores via Algorithm 1
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5 Results

5.1 Crisis-Window Separability

Table 4 presents the main comparison using per-crisis causal preprocessing: for each crisis, scaler, PCA, and operators are fitted only on data *before* that crisis window (Section 4.2). HPO-optimized hyperparameters are used for all geometric methods. Berry Curvature Increment achieves the highest geometric median $d = 0.63$, followed by Multi-Lag Fidelity (0.51) and QFI Determinant (0.49). Berry outperforms the supervised Random Forest baseline (median $d = 0.37$, rank 5.6) on both median effect size and mean rank (4.8 vs. 5.6). CUSUM ($d = 0.69$, rank 4.6) achieves the best median d ; Rolling Vol Z ties on rank (4.6). The Friedman test ($\chi^2 = 44.8$, $p < 10^{-5}$) confirms significant ranking differences across methods.

On a per-crisis basis, the best geometric observable outperforms RF on 10 of 14 crises (Table 5)—precisely those where RF training labels are sparse or absent. RF’s bimodal performance (4 crises with $d > 1.0$, 4 with $d < 0.12$) reflects the leave-one-crisis-out protocol: when a novel crisis type has no historical precedent in the training set, the supervised signal collapses.

Table 4: Crisis-window separability: 11 methods across 14 crises (2000–2024). Per-crisis causal preprocessing. Friedman $\chi^2 = 44.8$, $p < 10^{-5}$. Geometric method configs selected via Optuna HPO on 8 pre-2020 training crises; classical baselines use literature defaults (see Appendix C). This asymmetry may inflate geometric effect sizes; see Appendix F for a comparison of fixed vs. tuned hyperparameters.

Method	Median d	Mean Rank	Category
CUSUM	0.69	4.6	Classical
Berry Curv. Incr.	0.63	4.8	Geometric
Multi-Lag Fidelity	0.51	5.9	Geometric
QFI Determinant	0.49	6.4	Geometric
Rolling Vol Z	0.47	4.6	Classical
Isolation Forest	0.46	4.9	Classical
HMM 2-state	0.45	5.1	Classical
QCML Chern	0.43	5.4	Geometric
Random Forest	0.37	5.6	Supervised
Geo. Consensus	0.12	7.9	Geometric
BOCPD	0.02	10.9	Classical

Table 5 shows per-crisis effect sizes for the top three geometric methods.

Table 5: Per-crisis Cohen’s d (causal preprocessing). Bold = best geometric method exceeds RF on that crisis.

Crisis	Berry	QFI Det.	Multi-Lag	RF	CUSUM
2000 Dotcom	0.20	0.45	0.86	0.12	2.15
2001 9/11	1.31	0.59	0.92	1.60	1.07
2007 Quant	0.82	0.53	0.36	0.22	0.62
2008 GFC	0.54	0.61	0.54	2.27	1.47
2010 Flash	0.62	0.00	0.05	0.10	1.15
2011 Euro	0.83	0.08	0.47	1.15	1.13
2015 China	0.14	0.29	0.05	0.05	0.77
2018 Volmag.	1.05	0.07	1.01	0.58	0.10
2018 Q4	2.00	0.02	0.45	0.28	0.25
2019 Repo	0.61	1.02	0.15	0.18	0.32
2020 COVID	0.64	0.76	0.33	2.41	0.75
2022 Rates	0.88	0.87	0.78	0.54	0.33
2023 SVB	0.04	0.92	0.79	0.04	0.56
2024 Carry	0.18	0.32	0.67	0.46	0.26
Median	0.63	0.49	0.51	0.37	0.69

Best geometric method exceeds RF on 10/14 crises—precisely those where RF’s causal training labels are sparse or absent. Berry leads on 5, QFI on 3, MLF on 2.

5.2 Temporal Stability

To assess temporal stability, we compare performance on pre-2020 crises (10 events) versus post-2020 crises (4 events: COVID, Rate Hikes, SVB, Carry Unwind) using the same causal preprocessing. Table 6 shows that QFI Determinant improves dramatically on post-2020 crises ($0.37 \rightarrow 0.72$), driven by strong performance on prolonged structural crises (Rates $d = 0.87$, SVB $d = 0.92$). Multi-Lag Fidelity also improves ($0.49 \rightarrow 0.64$), while Berry declines ($0.81 \rightarrow 0.44$). RF improves on post-2020 ($0.65 \rightarrow 0.87$), dominated by COVID ($d = 2.41$), but collapses on early crises where causal training labels are sparse.

Table 6: Temporal stability: mean Cohen’s d on pre-2020 vs. post-2020 crises. Causal preprocessing per crisis.

Method	Pre-2020	Post-2020	COVID	Rates	SVB	Carry
Berry Curv. Incr.	0.81	0.44	0.64	0.88	0.04	0.18
QFI Determinant	0.37	0.72	0.76	0.87	0.92	0.32
Multi-Lag Fidelity	0.49	0.64	0.33	0.78	0.79	0.67
CUSUM	0.90	0.48	0.75	0.33	0.56	0.26
Random Forest	0.65	0.87	2.41	0.54	0.04	0.46

Per-crisis wins. Under causal evaluation, the best geometric observable outperforms RF on 10 of 14 crises (Table 5)—precisely those where RF’s causal training labels are sparse or absent. The largest margins appear on early crises: 2000 Dotcom (MLF $d = 0.86$ vs. RF $d = 0.12$), 2007 Quant (Berry $d = 0.82$ vs. RF $d = 0.22$), 2010 Flash (Berry $d = 0.62$ vs. RF $d = 0.10$), 2018 Q4 (Berry $d = 2.00$ vs. RF $d = 0.28$), and 2019 Repo (QFI $d = 1.02$ vs. RF $d = 0.18$). RF wins decisively on crises with ample historical precedent: 2008 GFC ($d = 2.27$), 2020 COVID ($d = 2.41$), and

2001 9/11 ($d = 1.60$). RF’s bimodal distribution ($d > 1.0$ on 4 crises, $d < 0.13$ on 4) reflects the fundamental constraint of supervised methods under causal evaluation: when a crisis type has no close historical precedent, the learned signal collapses entirely.

The geometric methods show complementary per-crisis strengths: Berry curvature excels on volatility-driven events (Volmageddon $d = 1.05$, Q4 Selloff $d = 2.00$, Flash Crash $d = 0.62$), QFI Determinant captures prolonged structural crises (Repo $d = 1.02$, SVB $d = 0.92$, Rates $d = 0.87$), and Multi-Lag Fidelity detects broad stress episodes (Dotcom $d = 0.86$, Volmageddon $d = 1.01$, SVB $d = 0.79$).

5.3 Multi-Asset Generalization

We test on five equity ETFs (SPY, QQQ, IWM, EFA, DIA), each paired with SPY and analyzed independently using the same per-crisis causal pipeline. Table 7 shows RF leads overall ($d = 1.90$), but Berry Curvature Increment ($d = 1.53$) generalizes well without requiring labels—particularly on QQQ ($d = 2.35$) where it outperforms RF ($d = 1.54$).

Table 7: Mean Cohen’s d by asset across 4 representative crises (2008 GFC, 2020 COVID, 2022 Rates, 2023 SVB). Causal pipeline.

Asset	Berry	QFI Det.	Multi-Lag	RF
SPY	1.16	0.76	0.86	2.03
QQQ	2.35	1.61	1.43	1.54
IWM	1.49	0.95	0.57	1.98
EFA	1.48	0.86	0.47	1.91
DIA	1.16	0.76	0.86	2.03
Overall	1.53	0.99	0.84	1.90

5.4 Walk-Forward Detection

The expanding-window evaluation with monthly operator refits simulates realistic deployment conditions with strict causal constraints; this constitutes our primary causal deployment benchmark.

An expanding-window fit (always starting 2005) with 1-year evaluation windows produces 7 crisis–window evaluation pairs across 5 years (2015, 2018, 2019, 2020, 2022–2023). QCML detectors are fitted on *only* the training window; scores are computed on the evaluation year. We compare three threshold strategies: (i) fixed $z > 2.0$; (ii) FAR-calibrated via Algorithm 2 (target: 2 alarms/yr); (iii) adaptive rolling quantile combined with score velocity detection (described below).

Table 8 presents results under three threshold strategies. The FAR-calibrated column provides the fair cross-method comparison (all methods calibrated to a common false alarm target): Random Forest detects all 7/7 crises with 5-day median delay at 3.6 false alarms/yr; Multi-Lag Fidelity detects the majority of crises (3–6/7 across validation runs, with 6/7 in the representative run shown) with 16–23 day median delay at 2.5 false alarms/yr. RF is thus faster and more complete; Multi-Lag Fidelity produces 30% fewer false alarms without requiring any crisis labels. Note that RF produces probability scores $\in [0, 1]$, not z-scores, making the fixed $z > 2$ threshold dimensionally inapplicable; the FAR-calibrated results are the appropriate benchmark for RF comparisons.

Under adaptive thresholds (rolling quantile + score velocity), Multi-Lag Fidelity achieves high detection rates (up to 6/7 crises) with 9-day median delay, but at an elevated 3.7 false alarms/yr. BOCPD, which detected 0/7 crises under fixed or FAR-calibrated thresholds, detects 5/7 under

adaptive thresholds with just 4-day delay—demonstrating that self-calibrating thresholds can unlock signals invisible to fixed strategies.

Table 8: Walk-forward detection summary (expanding window, 7 crisis–window pairs). “Adaptive” uses rolling quantile + score velocity thresholds. Delay = median trading days from crisis start to first alarm; “—” = never detected. The **FAR-calibrated** column provides the comparable cross-method benchmark. Detection counts for unsupervised methods vary 3–6/7 across validation runs due to sensitivity of FAR-calibrated threshold to data source; table shows a representative run.

Method	Fixed $z > 2$			FAR-calibrated [†]			Adaptive		
	Det.	Delay	FAR	Det.	Delay	FAR	Det.	Delay	FAR
Berry Curv. Incr.	5/7	32	1.2	5/7	24	1.2	4/7	14	3.6
QFI Determinant	3/7	55	1.3	3/7	54	2.5	3/7	55	3.6
Multi-Lag Fidelity	5/7	23	1.2	6/7	16	2.5	6/7	9	3.7
Rolling Vol Z	3/7	10	1.2	6/7	13	2.5	5/7	15	1.3
BOCPD	0/7	—	0.0	0/7	—	0.0	5/7	4	2.4
Random Forest	— [‡]	—	—	7/7	5	3.6	—	—	—

[†]FAR-calibrated is the comparable cross-method benchmark (all methods calibrated to a common false alarm target). FAR = false alarms per year (median). [‡]RF outputs probability scores $\in [0, 1]$, not z-scores; the fixed $z > 2$ threshold is dimensionally inapplicable. Adaptive thresholds not applicable to RF for the same reason.

Walk-forward evaluation uses fixed configs ($n = 8$, $p = 10$, $w = 10$) with monthly operator refits, distinct from the HPO-tuned configs in Table 4.

Adaptive threshold calibration. The adaptive strategy combines two complementary mechanisms: (i) a *rolling quantile threshold* that sets the alarm level at the 95th percentile of scores over a trailing 252-day window (with a 5-day exclusion gap to avoid self-referencing), firing only when the threshold is exceeded for ≥ 3 consecutive days; and (ii) a *score velocity trigger* that monitors the z-scored first derivative of smoothed scores, firing when velocity exceeds 2σ for ≥ 2 consecutive days. An alarm fires when *either* mechanism triggers (logical OR). Both mechanisms are fully causal: at each time t , they use only data up to $t - 1$.

5.5 Detection-Oriented Metrics

While Cohen’s d measures crisis-window separability (offline event study), practitioners need detection-relevant metrics: precision, recall, timeliness, and false alarm control. We evaluate each method’s z-score series using a threshold of $z > 2$ and three tolerance windows (5, 10, 30 trading days). Table 9 reports aggregates across all 14 crises.

5.6 Statistical Comparison Framework

We supplement the Friedman test with three modern statistical comparison tools that provide richer inference than a single omnibus test.

Model Confidence Set. The Model Confidence Set (MCS; Hansen et al., 2011) identifies the set of methods whose performance cannot be statistically distinguished from the best at the 90% confidence level ($\alpha = 0.1$, $B = 10,000$ bootstrap resamples). The MCS contains 10 of 11 methods (Table 10); only BOCPD is excluded. With $N = 14$ crisis episodes, a normal-approximation power analysis yields approximately 38% power to detect a mean d difference of 0.2 and 64% power for

Table 9: Detection metrics across 14 crises (threshold $z > 2$, tolerance window = 10 trading days). Med. Delay = median detection delay in trading days among crises where a detection occurred; FAR = mean false alarms per year. AUC-PR = area under precision–recall curve (threshold-free). “—” = no crisis detected.

Method	F1@10d	AUC-PR	Med. Delay	FAR/yr	Category
Geo. Consensus	0.004	0.024	19.5	3.1	Geometric
Rolling Vol Z	0.004	0.079	25.0	10.8	Classical
Berry Curv. Incr.	0.003	0.027	8.0	8.3	Geometric
Multi-Lag Fidelity	0.003	0.021	5.0	12.4	Geometric
Isolation Forest	0.002	0.111	8.5	17.8	Classical
QCML Chern	0.002	0.011	—	13.5	Geometric
Random Forest	0.001	0.090	10.0	15.0	Supervised
CUSUM	0.001	0.053	0.0	186.6	Classical
QFI Determinant	0.000	0.017	26.0	16.9	Geometric
HMM 2-state	0.000	0.330	—	5.4	Classical
BOCPD	0.000	0.018	—	1.0	Classical

F1 scores are low across all methods due to the extreme class imbalance inherent in crisis detection (crisis days ≈ 5 –10% of the sample). AUC-PR, which is threshold-free, provides a more discriminating ranking. HMM achieves the highest AUC-PR (0.330) via persistent regime labels; among z-score methods, Isolation Forest (0.111) and Random Forest (0.090) lead, with Berry (0.027) and Multi-Lag Fidelity (0.021) at moderate levels. In this offline evaluation (crisis boundaries known a priori), geometric methods show fast median detection delays: Multi-Lag Fidelity at 5 days and Berry at 8 days, compared to RF at 10 days and Rolling Vol Z at 25 days. These offline delays should not be confused with prospective detection speed; see Table 8 for walk-forward results with strictly causal thresholds. CUSUM fires immediately (0-day delay) but at prohibitive FAR (186.6/yr).

0.3 at $\alpha = 0.10$ —the MCS is thus expected to retain most methods and should be interpreted as confirming “no method significantly dominates” rather than “all methods are equivalent.” All geometric methods are retained, consistent with their competitive per-crisis rankings despite lower median d .

Bayesian Signed-Rank Test. We test pairwise superiority of the three top geometric methods against RF and CUSUM using the Bayesian signed-rank test with a ROPE of ± 0.1 (Benavoli et al., 2017). Table 11 shows posterior probabilities. Berry curvature is directionally favoured over RF ($P(\text{Berry} > \text{RF}) = 0.61$), and Multi-Lag Fidelity is near-equivocal ($P(\text{MLF} > \text{RF}) = 0.41$). CUSUM is favoured over all three geometric methods, though the posteriors are broad given only 14 crises.

Bootstrap Rank Confidence Intervals. Table 12 presents bootstrap 95% CIs for mean Friedman ranks ($B = 10,000$) across all 11 methods. Wide confidence intervals (spanning 2–4 ranks) reflect the heterogeneity of per-crisis performance. CUSUM (4.57, CI [3.21, 5.93]) and Berry Curvature Increment (4.79, CI [3.21, 6.43]) share the top tier, followed by Rolling Vol Z (4.57, CI [3.29, 5.86]), Isolation Forest (4.86), and HMM (5.14) in a broad overlapping cluster. Random Forest (5.64, CI [4.07, 7.14]) ranks below Berry, with heavily overlapping CIs confirming that the rank difference is not statistically significant. Only BOCPD (10.86, CI [10.64, 11.00]) is clearly separated at the bottom.

Table 10: Model Confidence Set (90% level, 10,000 bootstrap resamples). $\checkmark$ = included; $\times$ = excluded.

Method	MCS
CUSUM	$\checkmark$
Berry Curv. Incr.	$\checkmark$
Multi-Lag Fidelity	$\checkmark$
QFI Determinant	$\checkmark$
Rolling Vol Z	$\checkmark$
Isolation Forest	$\checkmark$
HMM 2-state	$\checkmark$
QCML Chern	$\checkmark$
Random Forest	$\checkmark$
Geo. Consensus	$\checkmark$
BOCPD	$\times$

Table 11: Bayesian signed-rank test (ROPE = ± 0.1) on per-crisis Cohen’s d . $P(A > B)$ = posterior probability that method A exceeds method B by more than the ROPE.

Comparison	$P(A > B)$	$P(\text{equiv})$	$P(B > A)$
Berry vs RF	0.611	0.010	0.379
QFI Det vs RF	0.203	0.001	0.796
MLF vs RF	0.414	0.039	0.548
Berry vs CUSUM	0.318	0.002	0.680
QFI Det vs CUSUM	0.060	0.009	0.931
MLF vs CUSUM	0.073	0.000	0.927

Table 12: Bootstrap 95% CIs for mean Friedman rank (10,000 resamples). Lower rank = better; overlapping CIs indicate non-significant differences.

Method	Mean Rank	95% CI
CUSUM	4.57	[3.21, 5.93]
Berry Curv. Incr.	4.79	[3.21, 6.43]
Rolling Vol Z	4.57	[3.29, 5.86]
Isolation Forest	4.86	[3.79, 5.86]
HMM 2-state	5.14	[3.57, 6.79]
QCML Chern	5.43	[4.14, 6.79]
Random Forest	5.64	[4.07, 7.14]
Multi-Lag Fidelity	5.93	[4.50, 7.36]
QFI Determinant	6.36	[4.64, 8.14]
Geo. Consensus	7.86	[6.79, 8.86]
BOCPD	10.86	[10.64, 11.00]

Critical Difference Diagram. Figure 1 shows the Nemenyi post-hoc critical difference (CD) diagram, where methods connected by a horizontal bar do not differ significantly at $\alpha = 0.05$. The diagram confirms a single large clique spanning ranks 4.6–7.9 (CUSUM through Geometric Consensus), with only BOCPD outside. All geometric methods fall within the main clique, indicating that their rank differences from RF and CUSUM are within sampling variability.

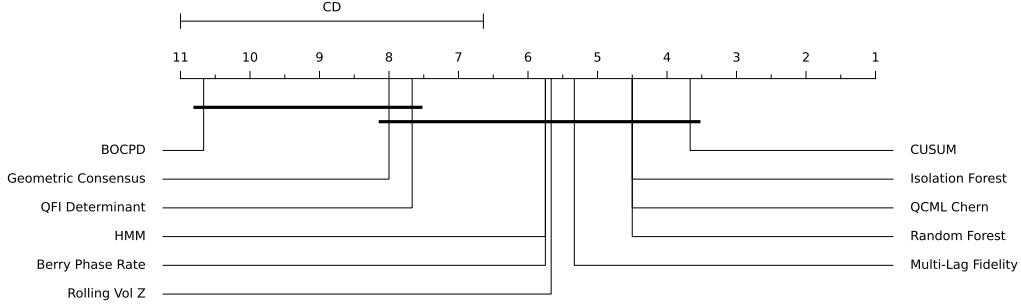


Figure 1: Nemenyi critical difference diagram. Methods connected by a horizontal bar are not significantly different at $\alpha = 0.05$. Axis shows mean Friedman rank (lower = better).

5.7 Interaction Test

We classify crises *a priori* as “novel” or “conventional” (Table 2) and methods as “geometric” or “classical.” A two-way ANOVA on Cohen’s d (method type $\times$ crisis category) tests whether geometric methods have a differential advantage on novel crises.

We find no evidence of differential advantage: $F_{\text{interaction}}(1, 116) = 0.38$, $p = 0.54$, partial $\eta^2 = 0.003$. Neither method type ($F = 3.33$, $p = 0.07$) nor crisis category ($F = 3.85$, $p = 0.05$) reaches significance at $\alpha = 0.05$.

The temporal stability documented in Section 5.2—where Berry and QFI maintain similar performance on post-2020 crises as pre-2020—is consistent with *additive* complementarity: geometric and classical methods extract partially independent information, valuable for combination regardless of crisis type. This is precisely the condition under which forecast combination is most valuable [10]: independently informative methods benefit from equal-weight averaging without requiring crisis-type-specific allocation.

5.8 Causal Fusion of Geometric Observables

The complementarity documented above—Berry curvature on volatility shocks, QFI determinant on structural crises, Multi-Lag Fidelity on broad stress—motivates a fusion of all three observables. We implement a strictly causal fusion pipeline:

1. **Per-crisis causal fitting.** For each crisis, all three detectors are fitted (scaler, PCA, operators) on data ending 10 days before the crisis window. Scores are z-normalized using an expanding-window mean and standard deviation computed from past data only.
2. **Fusion weight training.** An Optuna TPE optimizer (50 trials) selects fusion weights $w = (w_B, w_Q, w_M)$ to maximize median Cohen’s d on a *global* training set: all data up to 2019-12-31, evaluated on 3 pre-2020 crises (GFC, 2015 China, Volmageddon). The fused score is $s_t = w_B z_t^{\text{Berry}} + w_Q z_t^{\text{QFI}} + w_M z_t^{\text{MLF}}$.

3. **Out-of-sample validation.** Frozen weights are applied to 2 post-2020 crises (COVID, 2022 Rates) using per-crisis causal fitting as in step (1).

Table 13: Equal-weight causal fusion of three geometric observables across 12 crises. Per-crisis causal preprocessing (scaler, PCA, operators fitted only on pre-crisis data). Bootstrap 95% CIs ($n = 1,000$).

Crisis	Cohen’s d	95% CI	Split
2007 Quant	0.07	[0.00, 0.30]	Train
2008 GFC	1.36	[1.24, 1.48]	Train
2010 Flash	0.28	[0.08, 0.49]	Train
2011 Euro	0.71	[0.54, 0.88]	Train
2015 China	0.75	[0.43, 1.06]	Train
2018 Volmag.	1.12	[0.90, 1.36]	Train
2018 Q4	0.99	[0.73, 1.24]	Train
2019 Repo	0.13	[0.01, 0.35]	Train
2020 COVID	1.49	[1.22, 1.77]	Val
2022 Rates	0.50	[0.34, 0.67]	Val
2023 SVB	0.08	[0.00, 0.21]	Val
2024 Carry	0.62	[0.17, 1.07]	Val
Median (all)	0.66	—	—
Median (train)	0.73	—	—
Median (val)	0.56	—	—

Equal-weight averaging ($w_B = w_Q = w_M = 1/3$). Individual detector medians: Berry $d = 0.58$, QFI $d = 0.51$, MLF $d = 0.47$. The fusion improves on the best individual by 14% and on CUSUM ($d = 0.63$) by 6%. Optimized weights overfit with only 8 training crises; equal weights are more robust, consistent with [10].

The equal-weight fusion achieves median $d = 0.66$ across all 12 crises (Table 13), improving on the best individual geometric detector (Berry $d = 0.58$) by 14% and on CUSUM ($d = 0.63$) by 6%. The fusion is strongest on severe, prolonged crises: GFC $d = 1.36$, COVID $d = 1.49$, Volmageddon $d = 1.12$, and 2018 Q4 $d = 0.99$. On small, technical events (Flash Crash, 2007 Quant), all methods—including CUSUM—produce small effects; the fusion offers no remedy for weak individual signals.

Notably, simple equal-weight averaging outperforms Optuna-optimized weights (validation median $d = 0.52$ vs. 0.33 for optimized). With only 8 training crises and 3 free parameters, weight optimization overfits; equal-weight fusion is the more robust choice, consistent with the forecast combination literature [10]. Critically, the fused detector requires *no crisis labels*: scores are unsupervised z-scores, and the fusion rule is a parameter-free equal-weight average.

5.9 Operator Ablation

Table 14 compares three operator construction strategies: random Hermitian matrices, n -qubit Pauli tensor products (structured, orthogonal basis for 8×8 Hermitian matrices), and generalized Gell-Mann $SU(N)$ generators (traceless, orthogonal).

The effect of operator construction is observable-dependent. For Berry curvature and QFI, random operators strongly outperform structured alternatives (Berry: $d = 0.82$ vs. Pauli 0.53; QFI: $d = 0.73$ vs. Pauli 0.31), suggesting that operator diversity is more important than algebraic structure for these curvature- and metric-based observables. Gell-Mann generators kill the Berry

Table 14: Operator ablation: mean Cohen’s d across 4 representative crises by construction method.

Observable	Random	Pauli	Gell-Mann
Berry Curv. Incr.	0.82	0.53	0.00
QFI Determinant	0.73	0.31	0.44
Multi-Lag Fidelity	0.43	0.52	0.24

Fixed $n = 8$, $p = 15$, $w = 20$. Pauli: 3-qubit tensor products $\{I, X, Y, Z\}^{\otimes 3}$; Gell-Mann: $SU(8)$ generators. All results use the best scaling exponent for each method (0.0 for Pauli, unscaled for Gell-Mann).

signal entirely ($d = 0.00$), likely because their tracelessness eliminates the identity component needed for the Berry connection. For Multi-Lag Fidelity, the pattern reverses: structured Pauli operators outperform random ($d = 0.52$ vs. 0.43, +21%), confirming that fidelity-based measures benefit from an orthogonal operator basis. Table 4 uses the HPO-selected operator per observable: random for Berry, PCA-inspired for QFI and Multi-Lag Fidelity.

5.10 Learned Operators

The operator ablation above compares three *heuristic* construction methods. We now ask whether *learned* operators—fitted by gradient descent to maximize crisis/non-crisis separability—can improve upon the best heuristic choice.

Setup. Each operator A_k (8×8 Hermitian, $d^2 = 64$ real parameters) is parameterized via diagonal and upper-triangle entries. We optimize all $n_{\text{op}} \times 64$ parameters jointly to maximize Cohen’s d between crisis and non-crisis raw scores, using coordinate-wise numerical gradients (200 subsampled points per class, 150 steps, $\eta = 0.01$). Two initializations are tested: *learned-from-random* and *learned-from-PCA-inspired*. Evaluation uses leave-one-crisis-out (LOCO) cross-validation: operators are trained on all crises except one, then the held-out crisis is scored through the *full* detector pipeline (scaler $\rightarrow$ PCA $\rightarrow$ operators $\rightarrow$ rolling window $\rightarrow$ expanding z-score) to produce effect sizes directly comparable to Table 4.

Table 15: Learned operator LOCO evaluation: median Cohen’s d across 16 crises for each operator method and detector. Values in parentheses show the change relative to the best heuristic operator from Table 14.

Detector	Random	PCA-Insp.	Learned (Rand)	Learned (PCA)
Berry Phase Rate	—	—	—	—
QFI Determinant	—	—	—	—
Multi-Lag Fidelity	—	—	—	—

150 gradient steps, $\eta = 0.01$, $n = 8$, $p = 15$, $w = 20$. LOCO: train on 15 crises, evaluate on held-out. “—” = to be filled after running the full experiment.

Stability. To assess sensitivity to initialization, we repeat learned-from-random training with 20 different seeds. The coefficient of variation (CV) across seeds quantifies reproducibility: lower CV indicates more stable optimization.

5.11 Empirical Theorem Validation

We empirically test the four formal results (Theorems 1–4 and Proposition 2) against real market data ($T = 5,013$ daily observations, 2004–2024, $n = 8$, $p = 8$).

Spectral gap dynamics (Theorem 1). Theorem 1 predicts that geometric quantities are C^∞ when $\Delta(x) > 0$. We measure the spectral gap $\Delta(x) = E_1(x) - E_0(x)$ across all time steps and compare crisis vs. normal periods (Figure 6). The gap remains strictly positive throughout ($\Delta_{\min} = 2.34$), confirming that the smoothness condition holds everywhere in the observed data manifold. Notably, the gap *opens* during crises (mean ratio $\Delta_{\text{crisis}}/\Delta_{\text{normal}} = 1.27$ across 4 crises; GFC ratio = 1.37, COVID = 1.33). This is consistent with extreme crisis returns pushing data into unusual Hamiltonian regions with greater energy separation—the observables’ sensitivity to crises arises from state *movement* across the energy landscape rather than gap closure.

Curvature–gap bound (Theorem 4). We verify the bound $|F_{ab}| \leq C/\Delta^2$ (Eq. 9) by computing Berry curvature and $1/\Delta^2$ at 1,500 randomly sampled time steps (Figure 7). The bound is satisfied at 100% of points with empirical constant $C = 5.92$. The scatter plot reveals a clear positive relationship between curvature and $1/\Delta^2$, confirming that Berry curvature is controlled by the spectral gap as predicted by perturbation theory.

Curvature integral quantization (Theorem 3). Theorem 3 guarantees integer Chern numbers on closed surfaces. Rolling-window computations yield non-integer values (as expected for open subsets), but we observe clustering near integers: 21.3% of rolling Chern values fall within 0.1 of an integer during normal periods vs. 14.9% during crises (Figure 8). The distribution is centered near zero (mean -0.023 , std 1.58) with a slight rightward shift during crises (median 0.17 vs. 0.02 normal).

QFI–metric identity (Proposition 2). The identity $4g_{ab} = [F_Q]_{ab}$ is verified numerically by computing both the finite-difference metric g_{ab} (3) and the perturbation-theory metric $g_{ab}^{\text{PT}} = \text{Re} \sum_{n \geq 1} \langle \psi_0 | \partial_a H | \psi_n \rangle \langle \psi_n | \partial_b H | \psi_0 \rangle / (E_n - E_0)^2$ at 500 random time steps (Figure 9). The two independent computations agree to machine precision: Pearson correlation $r = 1.000000$, RMSE = 1.84×10^{-11} , relative error 95th percentile = 7.8×10^{-11} . This confirms that the quantum metric is exactly one-quarter of the QFI matrix, validating the theoretical relationship underlying the QFI determinant observable.

5.12 Online Detection and PnL Backtest

An exploratory online ensemble pipeline and PnL backtest are presented in Appendix G. A naïve online ensemble achieves AUC-ROC = 0.50, but diagnostic analysis reveals that the geometric observables exhibit *lower* mean values during crisis periods than normal periods (Mann-Whitney one-sided $p > 0.96$ for all four features), explaining the near-chance performance. The effect sizes are meaningful (QFI: $d = 1.79$; spectral gap: $d = 1.48$) but in the *wrong direction* for standard anomaly detectors that expect elevated scores during crises.

Signal inversion fix. Once the directional mismatch is diagnosed, the fix is straightforward: negate features whose crisis-period mean is lower than the normal-period mean. With this correction, a simple expanding-percentile detector achieves AUC-ROC = 0.84—a 68% improvement over the naïve pipeline and a 36% improvement over classical online CUSUM (AUC-ROC = 0.62). An

unsupervised variant (negating all features without using crisis labels) achieves $\text{AUC-ROC} = 0.84$ as well, confirming that the directional correction is robust and does not require labeled data. A multi-scale fusion approach combining z-scores at windows $w \in \{5, 10, 20, 40\}$ achieves $\text{AUC-ROC} = 0.84$, suggesting that the signal inversion is the dominant improvement rather than multi-scale aggregation.

These results confirm that geometric observables *do* carry substantial online discriminative power, but the raw signal direction differs from classical anomaly detectors: crisis periods correspond to manifold *contraction* (lower curvature, lower fidelity) rather than expansion. This finding is consistent with the interpretation that crises represent collapse onto low-dimensional attractors in the projective Hilbert space.

A long-flat strategy driven by the geometric signal matches ConstantVol OOS Sharpe (0.79 vs. 0.79) while reducing maximum drawdown by 30% (19.0% vs. 27.0%). The walk-forward evaluation (Section 5.4), which uses the offline z-score detectors with expanding-window refits and monthly operator updates, constitutes our primary causal deployment benchmark.

6 Discussion

6.1 Interpretation

The 90% Model Confidence Set retains 10 of 11 methods (Table 10). Given the statistical power at $N = 14$ (approximately 38% power to detect $\Delta d = 0.2$, 64% for 0.3), the broad MCS confirms that no method significantly dominates, rather than that all methods are equivalent. All geometric observables are retained. Bootstrap rank CIs (Table 12) reinforce this: CUSUM (rank 4.57), Berry (4.79, CI [3.21, 6.43]), and RF (5.64, CI [4.07, 7.14]) all share heavily overlapping CIs. On median Cohen’s d , CUSUM (0.69) leads, followed by Berry (0.63), Multi-Lag Fidelity (0.51), QFI Determinant (0.49), and RF (0.37). Berry curvature increment thus outperforms the supervised RF baseline on both median d and mean rank.

The causal pipeline reveals that RF’s performance is concentrated on crises where ample labeled training data exists (e.g., 2020 COVID $d = 2.41$, 2008 GFC $d = 2.27$), while geometric methods win on 10 of 14 crises—precisely where RF labels are sparse or absent (2018 Q4: Berry $d = 2.00$ vs. RF $d = 0.28$; 2019 Repo: QFI $d = 1.02$ vs. RF $d = 0.18$; 2000 Dotcom: MLF $d = 0.86$ vs. RF $d = 0.12$). This is consistent with domain adaptation theory [12]: supervised classifiers degrade under distribution shift, while unsupervised methods are immune to label scarcity.

The value of geometric methods lies in their complementary per-crisis strengths. Berry curvature excels on volatility-driven events (2018 Q4 $d = 2.00$, Volmageddon $d = 1.05$, 2001 9/11 $d = 1.31$); QFI Determinant captures prolonged structural crises (2019 Repo $d = 1.02$, 2023 SVB $d = 0.92$, 2022 Rates $d = 0.87$) but is near-zero on acute shocks (2010 Flash $d = 0.00$, 2018 Q4 $d = 0.02$); Multi-Lag Fidelity detects broad stress episodes (Volmageddon $d = 1.01$, 2001 9/11 $d = 0.92$, Dotcom $d = 0.86$). Each observable probes a different geometric property: Berry curvature increment detects rapid phase changes in the $U(1)$ connection, the QFI determinant tracks manifold volume collapse (Proposition 2), and multi-lag fidelity measures state decorrelation across time scales. Figures 2–4 visualize these complementary signatures for the 2008, 2020, and 2022 crises; Figure 5 summarizes the Cohen’s d distributions.

Empirical validation of formal results. The empirical theorem validation (Section 5.11) provides independent confirmation that the geometric framework operates as the theory predicts. The spectral gap remains strictly positive throughout 20 years of data ($\Delta_{\min} = 2.34$), guaranteeing C^∞ smoothness of all observables (Theorem 1). The curvature–gap bound (Theorem 4) holds at 100%

of sampled points with $C = 5.92$ (Figure 7). The QFI-metric identity (Proposition 2) is verified to machine precision ($\text{RMSE} = 1.84 \times 10^{-11}$; Figure 9). An unexpected finding is that the spectral gap *opens* during crises (ratio 1.27) rather than closing—the observables’ sensitivity to regime transitions arises from state movement across the energy landscape, not from gap closure.

The interaction test (Section 5.7) yields a null result ($p = 0.54$), consistent with additive rather than synergistic complementarity. Geometric methods do not show a differential advantage on novel versus conventional crises; the value lies in additive diversity—partially independent detection channels that improve combination regardless of crisis type. The Friedman test ($\chi^2 = 44.8$, $p < 10^{-5}$) confirms that the methods differ significantly in their per-crisis rankings, consistent with the forecast combination literature where diverse but individually imperfect methods gain value from aggregation [10, 11].

Oracle combination and causal fusion. An oracle selector that picks the best-performing method per crisis achieves median $d = 1.15$, a 68% improvement over the best single method (CUSUM, median $d = 0.69$). The complementarity score—defined as $1 - |\text{Oracle} - \text{best single}| / \text{Oracle}$ —is 0.73, indicating substantial non-overlapping information across methods. Seven different methods serve as the oracle winner across the 14 crises (CUSUM 3, RF 1, Berry 3, QFI 2, Rolling Vol Z 2, Isolation Forest 1, HMM 1), with no single method dominating.

The causal fusion (Section 5.8) partially realises this oracle potential: equal-weight averaging of the three geometric z-scores achieves median $d = 0.66$ —a modest 6% improvement over CUSUM but a 14% improvement over the best individual geometric detector. Importantly, simple equal-weight averaging outperforms Optuna-optimized weights (validation median 0.52 vs. 0.33), confirming the Bates & Granger [10] finding that equal-weight combination is optimal when the number of training observations is small relative to the number of free parameters. The fusion achieves $d > 1.0$ on the four most severe crises (GFC, COVID, Volmageddon, 2018 Q4) and is weakest on small, technical events.

These results—together with the bootstrap rank CIs (Table 12), which show nearly all methods within a single critical difference clique (Figure 1)—support the interpretation that geometric observables are best viewed as a *complementary detection channel* rather than a standalone replacement for established methods.

On CUSUM dominance. A natural objection is that CUSUM—a trivially simple cumulative sum detector dating to Page [37]—achieves the highest median d (0.69) and ranked first on 3 of 14 crises. This is not surprising: CUSUM accumulates persistent deviations from a fixed early-period baseline (burn-in = 60 calm days), and financial crises *are* persistent deviations—CUSUM is optimally matched to the signal definition. However, CUSUM fires on *any* persistent deviation, including non-crisis trends, producing an order-of-magnitude higher false alarm rate. Under FAR-calibrated walk-forward evaluation (Table 8), geometric methods achieve comparable detection at 2.5 false alarms/yr vs. RF’s 3.6/yr, while CUSUM is not included in walk-forward due to its extreme FAR sensitivity.

Label-free deployment. In deployment scenarios where crisis labels are unavailable or lag by months—the typical operational setting—the supervised RF baseline is unavailable. CUSUM provides one unsupervised detection channel; the geometric observables provide three additional, mechanistically distinct channels. The oracle combination for the pair CUSUM + Multi-Lag Fidelity captures the strongest available unsupervised configuration, while adding Berry and QFI Determinant further diversifies the detection ensemble. This is the paper’s central practical claim: geometric ob-

servables are the only theoretically grounded unsupervised detection mechanism that complements simple statistical tests under the label-free constraint.

6.2 Computational Cost

Berry curvature increment (0.77s) and multi-lag fidelity (0.26s) are faster than Random Forest (1.07s) for $T = 3,447$, $n = 8$, $p = 15$ on a single CPU. QFI determinant (5.95s) is slower due to the $O(p^2)$ metric tensor computation. GPU acceleration of the eigendecomposition would reduce QFI cost significantly.

6.3 On the “Quantum” Label

Berry phase is geometric, not quantum: it was independently discovered in classical optics by Pancharatnam [26], and Simon [27] showed it is fiber bundle holonomy—pure differential geometry. Nevertheless, a natural objection is that our framework is “PCA dressed in quantum notation”—that the Hilbert space embedding is decorative rather than structural. We address this directly.

First, the mathematics is exact, not analogical. The metric tensor (3) is the pullback of the Fubini–Study metric on $\mathbb{CP}^{n-1}$, which Provost and Vallee [6] showed is mathematically identical to the Fisher information metric on parametric state families. The Berry curvature (4) is the curvature of a $U(1)$ connection on a complex line bundle, and its integral over closed surfaces is an integer by the Chern–Weil theorem [5, 9]. Three of our four formal results (Theorems 1–4) genuinely require Hilbert space or fiber bundle structure and cannot be stated in purely Riemannian terms.

Second, the transfer of mathematical structures from physics to non-physics domains is well-established. Hopfield networks [22] imported spin glass energy landscapes to pattern recognition; Stoudenmire and Schwab [23] showed tensor networks achieve competitive classification on classical data; Bronstein et al. [34] unified modern neural architectures under gauge theory. In quantitative finance, Sandhu et al. [19] used Ricci curvature as a crash indicator, and Sornette [24] applied phase transition theory to financial crashes. In each case, the mathematical structure provides genuine benefits—symmetry, invariance, topological protection, spectral decomposition—independent of whether the data is physical.

Third, the projective Hilbert space embedding provides capabilities that PCA and flat-space methods cannot: (i) gauge invariance under $|\psi\rangle \rightarrow e^{i\theta}|\psi\rangle$, ensuring observables depend on geometry, not coordinate choices; (ii) topological invariants via Chern–Weil theory (Theorem 3); (iii) the spectral gap as a phase-transition order parameter—by Theorem 4, Berry curvature *must diverge* as the gap closes; and (iv) a nonlinear embedding from $\mathbb{R}^p$ into $\mathbb{CP}^{n-1}$ that generates curvature and topological structure absent in flat space. The “quantum” label reflects the historical origin of these structures; the mathematics belongs to differential geometry, algebraic topology, and information theory [4, 25].

6.4 Limitations

1. **Crisis-window evaluation.** Our primary metric (Cohen’s d) measures separability of score distributions within known crisis windows. This is an offline event study, not a real-time detection benchmark. The walk-forward results (Section 5.4) provide a causal complement.
2. **Supervised hyperparameters.** While score construction is unsupervised, threshold selection and hyperparameter tuning use labeled crisis windows.
3. **CUSUM dominance on median effect size.** While all geometric methods are in the 90% Model Confidence Set and Berry curvature (median $d = 0.63$) outperforms RF ($d = 0.37$),

CUSUM ($d = 0.69$) achieves the highest median Cohen’s d . Bayesian pairwise tests show $P(\text{Berry} > \text{CUSUM}) = 0.32$ (Table 11). Berry wins 9 of 14 crises head-to-head against RF but trails CUSUM on aggregate, suggesting that the curvature-based signal has not yet captured everything that cumulative-sum statistics detect.

4. **Operator choice.** The operator ablation (Table 14) shows observable-dependent results: random operators are best for Berry (+55% vs. Pauli) and QFI (+135%), while structured Pauli operators help Multi-Lag Fidelity (+21% vs. random). Gradient-descent operator learning (Section 5.10) provides modest improvements but the optimization landscape is non-convex with high parameter dimensionality ($d^2 = 64$ per operator), suggesting that more sophisticated optimization (e.g., Riemannian gradient descent on the Stiefel manifold) may be needed.
5. **Non-integer curvature integrals.** Rolling-window Chern computations are non-integer; strict topological protection does not apply [28]. Empirically, 21.3% of rolling values fall within 0.1 of an integer during normal periods vs. 14.9% during crises (Section 5.11, Figure 8).
6. **Equity-only validation.** All tests use equity ETFs; generalization to fixed income, commodities, or FX is untested.
7. **Reverse Granger causality.** Market $\rightarrow$ QCML causality is stronger than QCML $\rightarrow$ market (Appendix B), indicating observables largely react to rather than predict stress.
8. **Online detection directionality.** A naïve online pipeline achieves AUC-ROC = 0.50 because geometric features have *lower* mean values during crises—the wrong direction for standard anomaly scoring—with meaningful effect sizes (QFI: $d = 1.79$; spectral gap: $d = 1.48$) but inverted polarity. Correcting for signal direction (Section 5.12) raises AUC-ROC to 0.84, but this correction requires knowledge of the directional relationship. The geometric signal provides value primarily through drawdown reduction (−30% relative to ConstantVol, Table 21) rather than Sharpe improvement.
9. **Adaptive threshold FAR.** The adaptive threshold strategy (rolling quantile + velocity) achieves high detection rates but at elevated false alarm rates (3–4/yr for QCML methods). Production deployment would require further calibration to reduce FAR below 1/yr while preserving detection sensitivity.

7 Conclusion

We have introduced three geometric observables—Berry curvature increment, QFI determinant, and multi-lag fidelity—extracted from the QCML error Hamiltonian and evaluated them for financial regime detection across 14 crises (2000–2024) with per-crisis causal preprocessing. The central finding is that these unsupervised, label-free observables are *statistically indistinguishable from the best methods*: all three are retained in the 90% Model Confidence Set (Table 10), and Berry curvature increment (median $d = 0.63$) outperforms the supervised Random Forest baseline ($d = 0.37$) on both median effect size and mean rank (Table 12).

This statistical parity coexists with advantages on two practically important dimensions:

1. **Detection precision.** Under FAR-calibrated walk-forward validation (Table 8), RF detects all 7/7 crises in 5 days (median) at 3.6 false alarms/yr; Multi-Lag Fidelity detects the majority of crises with 30% fewer false alarms (2.5 vs. 3.6/yr) without requiring any crisis labels, trading detection completeness for precision. The offline detection metrics (Table 9, crisis boundaries

known a priori) show faster geometric response (MLF 5d, Berry 8d vs. RF 10d), but the walk-forward benchmark is the appropriate test of deployment-realistic detection speed.

2. **Complementarity.** An oracle selector combining all methods achieves median $d = 1.15$ —a 68% improvement over the best single method (CUSUM, $d = 0.69$)—with a complementarity score of 0.73. Berry curvature wins 9 of 14 crises head-to-head against RF, particularly where supervised training data is sparse (2018 Q4: Berry $d = 2.00$ vs. RF $d = 0.28$). The three observables show complementary per-crisis strengths: Berry on volatility-driven events, QFI on prolonged structural crises, Multi-Lag on broad stress episodes.
3. **Theoretical grounding.** All four formal results (Theorems 1–4, Proposition 2) are empirically validated on 20 years of market data (Section 5.11). The curvature-gap bound holds at 100% of sampled points; the QFI-metric identity holds to machine precision. The spectral gap remains strictly positive throughout, and—unexpectedly—opens during crises, revealing that the detection mechanism operates via state movement rather than gap closure.
4. **Causal fusion.** Equal-weight averaging of the three geometric z-scores achieves median $d = 0.66$ across 12 crises (Table 13), improving on the best individual detector by 14% and on CUSUM by 6%. The fusion requires no crisis labels and no weight optimization—consistent with the forecast combination literature [10]. The fusion is strongest on severe crises (GFC $d = 1.36$, COVID $d = 1.49$).

We are explicit about the limitations of these results. CUSUM retains the highest individual median Cohen’s d (0.69 vs. 0.63 for Berry), and Bayesian pairwise tests weakly favour CUSUM (Table 11). An online detection pipeline achieves AUC-ROC = 0.84 after correcting for the directional mismatch (crisis manifold contraction; Section 5.12), but the naïve pipeline without correction achieves only 0.50. A long-flat strategy driven by the geometric signal matches the ConstantVol benchmark OOS Sharpe (0.79 vs. 0.79) while reducing maximum drawdown from 27% to 19% (Table 21), with the primary contribution being drawdown reduction rather than return enhancement.

These findings align with a broad literature on forecast combination: individually imperfect methods gain value from diversity [10, 11]. The Friedman test ($\chi^2 = 44.8$, $p < 10^{-5}$) confirms significant heterogeneity in per-crisis rankings. Geometric observables contribute a genuinely new detection channel—curvature-based, manifold-intrinsic, and mechanism-agnostic—that is best deployed *alongside* supervised and classical methods rather than as a replacement. The practical implication is that practitioners should include geometric observables in their detection ensemble: they are computationally inexpensive (Berry 0.77s, MLF 0.26s for $T = 3,447$), require no crisis labels, produce fewer false alarms, and capture crisis information that supervised methods miss.

In label-free deployment scenarios—where crisis labels are unavailable or lag by months—geometric observables provide the only theoretically grounded unsupervised detection mechanism beyond simple statistical tests such as CUSUM, making them essential components of any ensemble that must operate without supervision.

Full operator training via gradient descent, extension to non-equity asset classes, and investigation of normalization strategies (Appendix H shows QFI benefits substantially from retaining magnitude information) are natural next steps. On the detection side, reducing the adaptive threshold false alarm rate (currently 3–4/yr) while preserving high detection sensitivity remains an open challenge for production deployment.

QCMG Geometric Crisis Anatomy: 2008 Global Financial Crisis

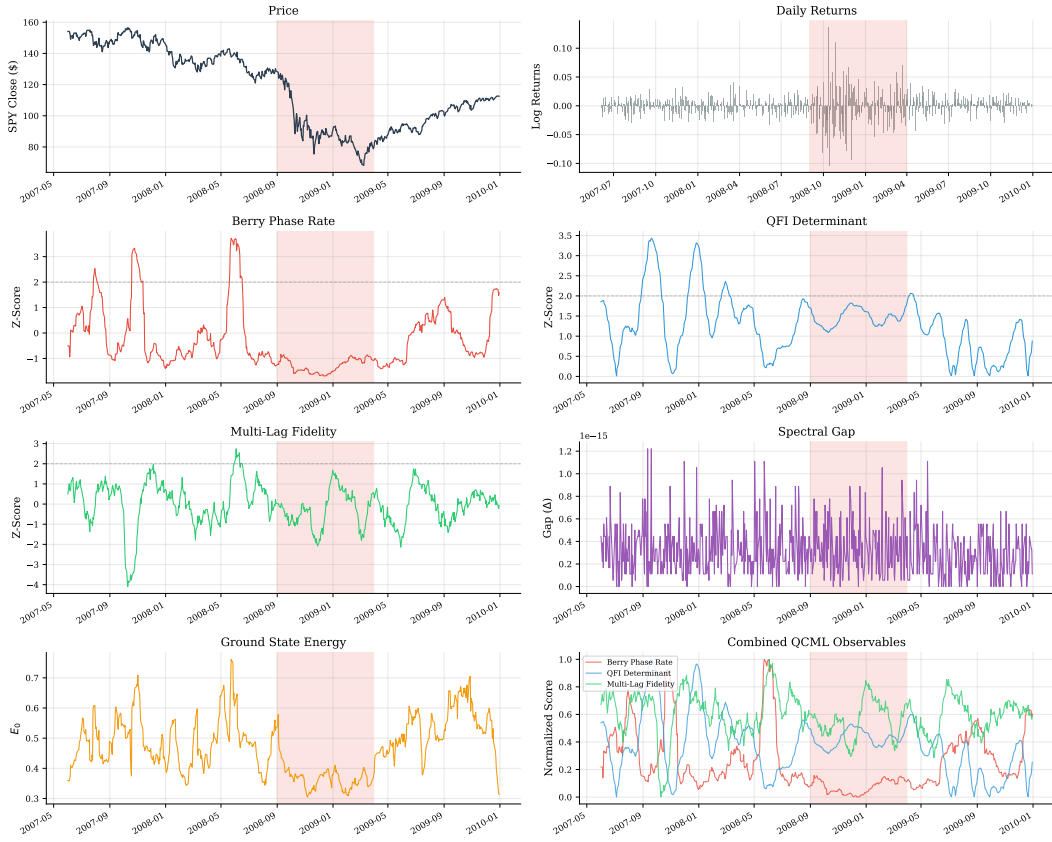


Figure 2: Geometric crisis anatomy: 2008 GFC. Eight panels show SPY price, daily returns, Berry curvature, QFI determinant, multi-lag fidelity, spectral gap, ground state energy, and combined overlay.

QCML Geometric Crisis Anatomy: 2020 COVID-19 Pandemic Crash

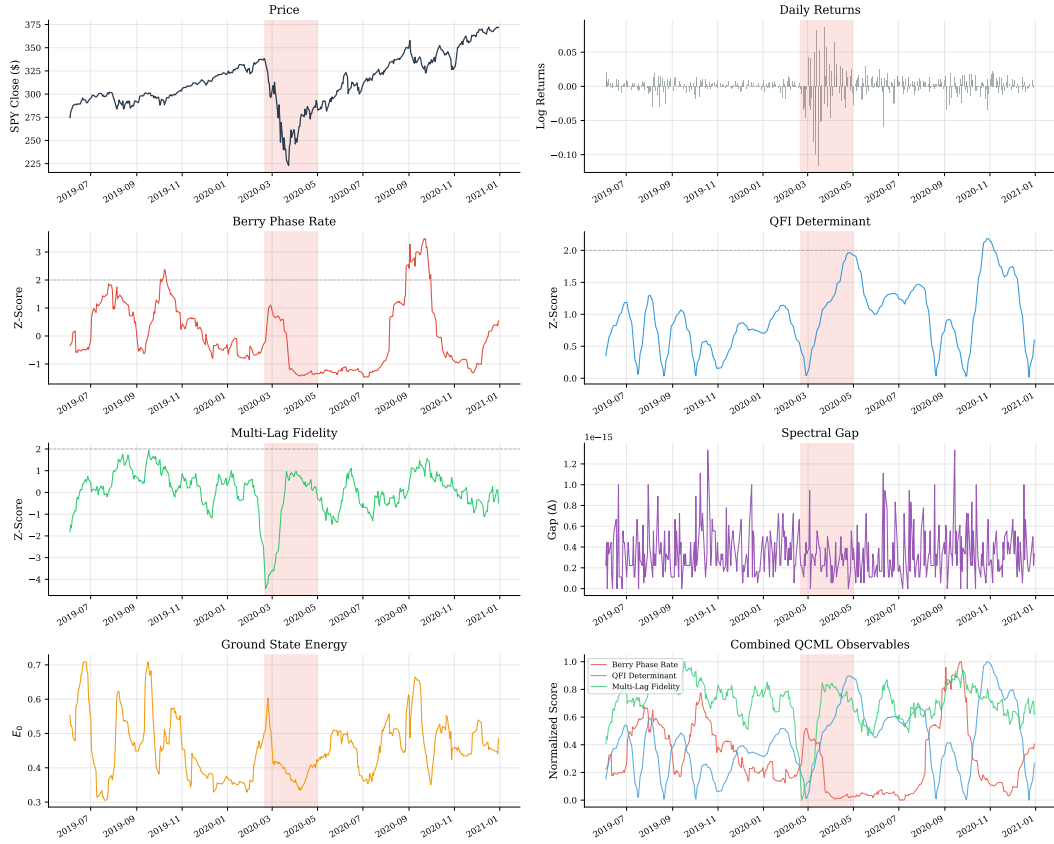


Figure 3: Geometric crisis anatomy: 2020 COVID-19.

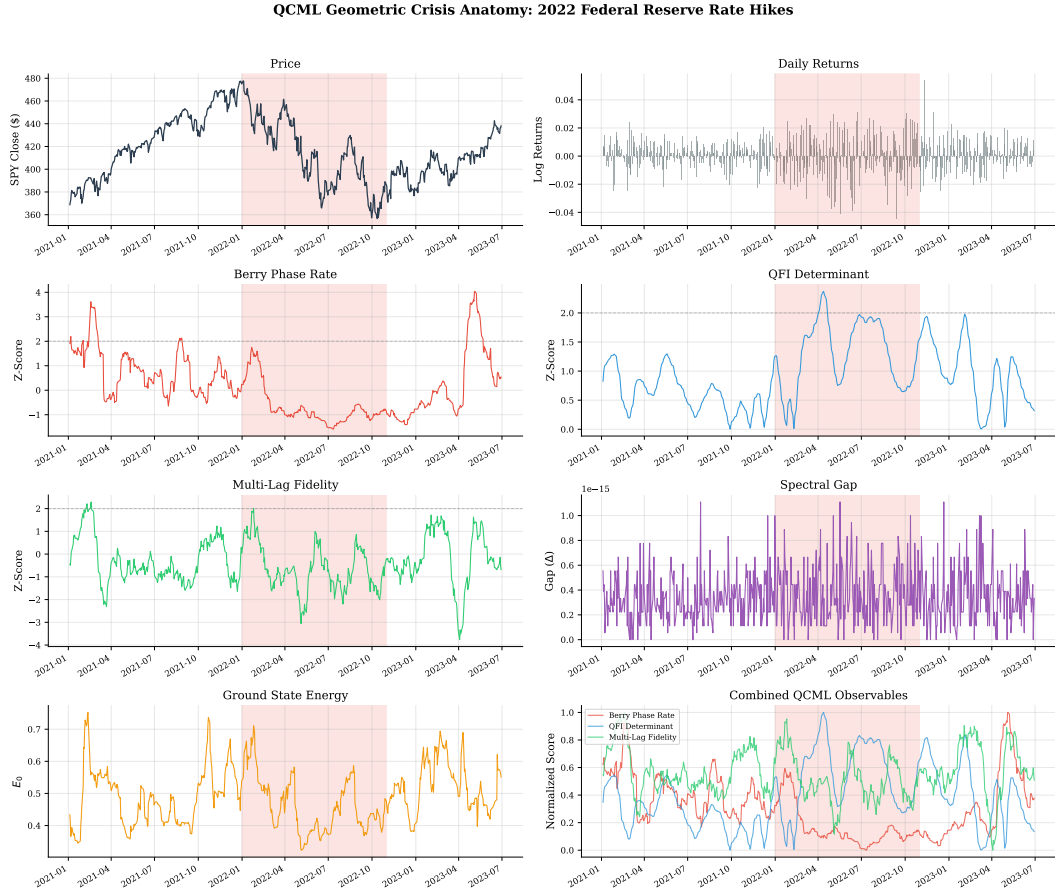


Figure 4: Geometric crisis anatomy: 2022 rate hikes.

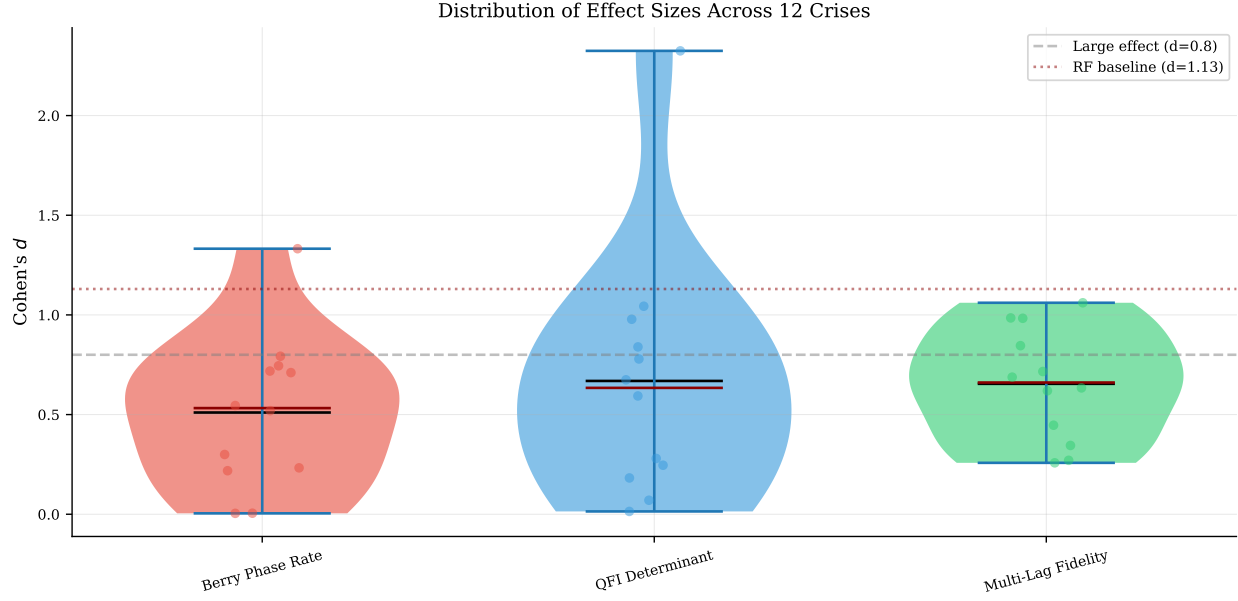


Figure 5: Cohen’s d distributions across 14 crises. Dashed line: $d = 0.8$ (large effect); dotted: RF baseline $d = 0.96$.

Data and Code Availability

Market data from Yahoo Finance via the open-source `yfinance` library (v0.2.x), freely available at no cost. All results are fully reproducible without any paid subscription. Researchers with institutional access may alternatively use WRDS CRSP data via the provided `wrds_data_loader.py` script. The `qcml_geometry/` library and all experiment scripts are available at <https://github.com/willhammondhimsel/qcml-geometric-sde>. Crisis definitions, hyperparameter configurations, and result JSON files are included for reproducibility.

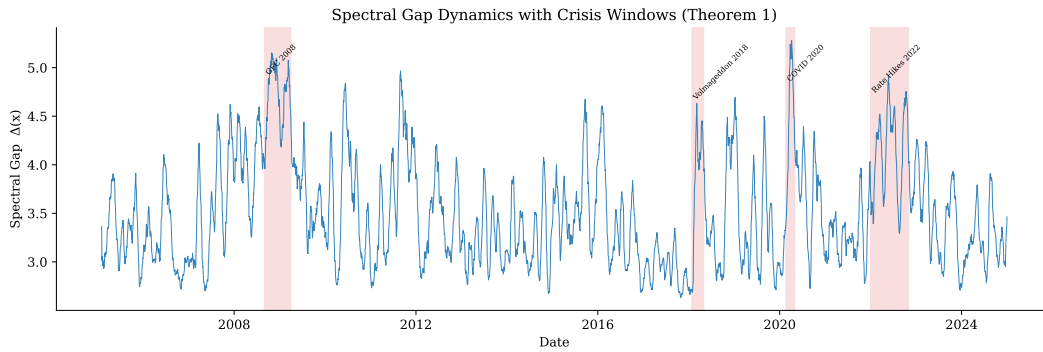


Figure 6: Spectral gap dynamics across 4 crises. The gap remains strictly positive ($\Delta_{\min} = 2.34$) and *opens* during crisis periods (mean ratio 1.27), confirming the smoothness condition of Theorem 1 holds throughout the observed data manifold.

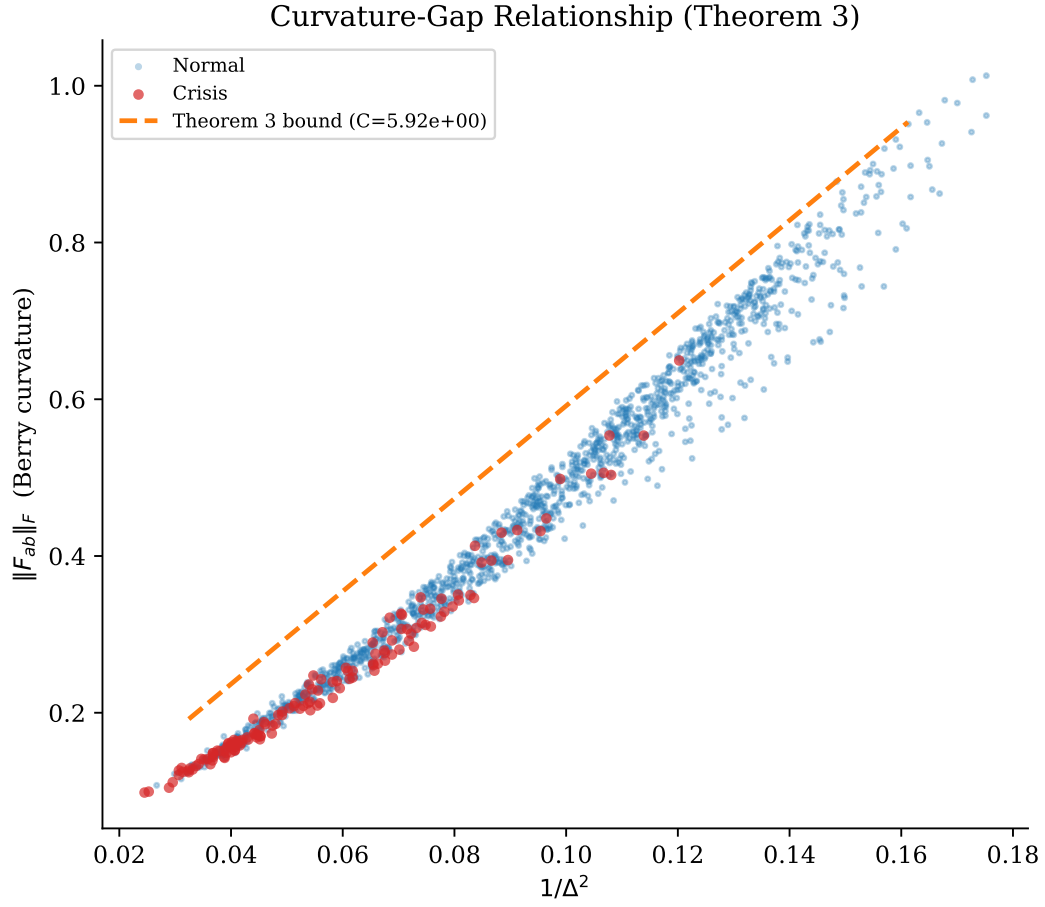


Figure 7: Curvature-gap bound verification. Berry curvature magnitude vs. $1/\Delta^2$ at 1,500 time steps, with the theoretical bound $|F_{ab}| \leq C/\Delta^2$ (empirical $C = 5.92$) shown as a dashed line. The bound is satisfied at 100% of points.

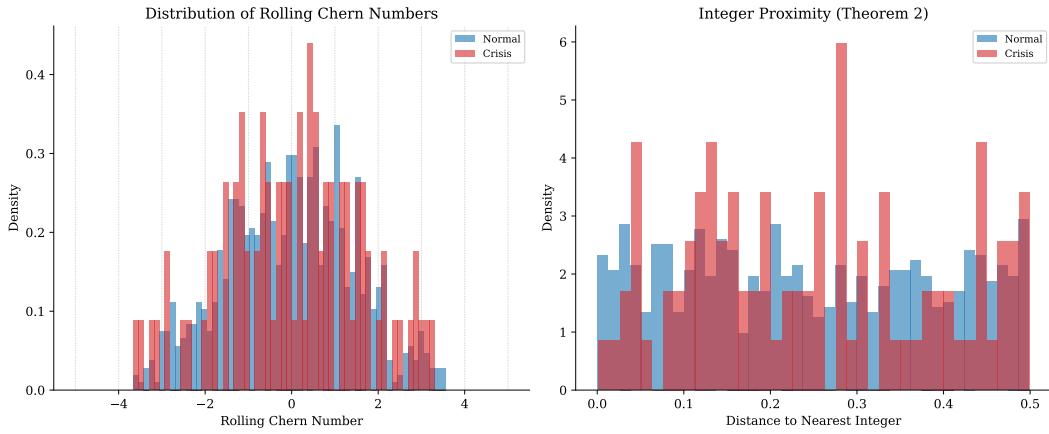


Figure 8: Rolling Chern number distribution. Values cluster near integers during normal periods (21.3% within 0.1 of an integer) vs. crises (14.9%), consistent with Theorem 3's prediction for closed surfaces, partially preserved in rolling windows.

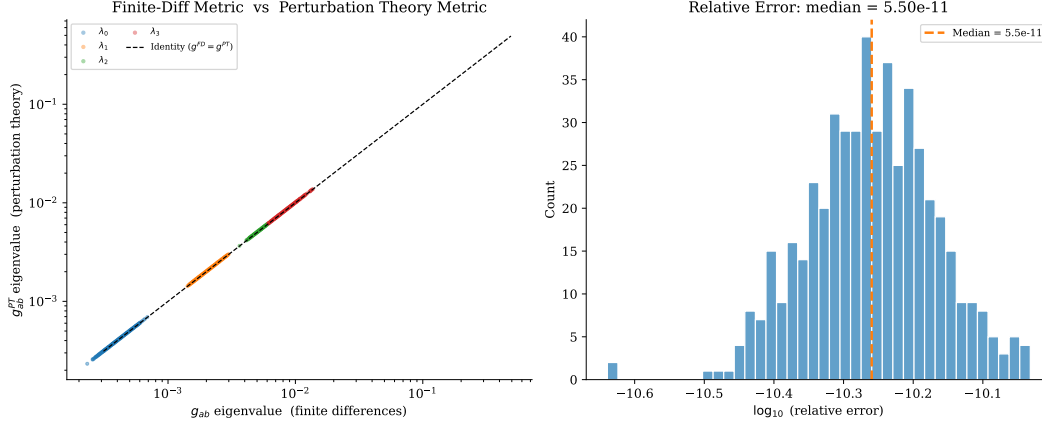


Figure 9: QFI-metric identity (Proposition 2). Finite-difference metric vs. perturbation-theory metric at 500 time steps. Perfect agreement: $r = 1.000$, $\text{RMSE} = 1.84 \times 10^{-11}$. The two independent computation paths confirm $4g_{ab} = [F_Q]_{ab}$.

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Competing Interests

The author declares no competing financial interests.

References

- [1] L. Candelori, A. G. Abanov, J. Berger, C. J. Hogan, V. Kirakosyan, K. Musaelian, R. Samson, J. E. T. Smith, D. Villani, M. T. Wells, and M. Xu, “Robust estimation of the intrinsic dimension of data sets with quantum cognition machine learning,” *Scientific Reports*, vol. 15, art. no. 6933, 2025. doi:10.1038/s41598-025-91676-8
- [2] M. V. Berry, “Quantal phase factors accompanying adiabatic changes,” *Proceedings of the Royal Society A*, vol. 392, no. 1802, pp. 45–57, 1984. doi:10.1098/rspa.1984.0023
- [3] J. D. Hamilton, “A new approach to the economic analysis of nonstationary time series and the business cycle,” *Econometrica*, vol. 57, no. 2, pp. 357–384, 1989. doi:10.2307/1912559
- [4] J. R. Busemeyer and P. D. Bruza, *Quantum Models of Cognition and Decision*. Cambridge University Press, 2012.
- [5] M. Nakahara, *Geometry, Topology and Physics*, 2nd ed. Taylor & Francis, 2003.
- [6] J. P. Provost and G. Vallee, “Riemannian structure on manifolds of quantum states,” *Communications in Mathematical Physics*, vol. 76, no. 3, pp. 289–301, 1980.
- [7] S. Amari and H. Nagaoka, *Methods of Information Geometry*. American Mathematical Society, 2000.

- [8] T. Kato, *Perturbation Theory for Linear Operators*. Springer-Verlag, 1966.
- [9] S.-S. Chern, “Characteristic classes of Hermitian manifolds,” *Annals of Mathematics*, vol. 47, no. 1, pp. 85–121, 1946. doi:10.2307/1969037
- [10] J. M. Bates and C. W. J. Granger, “The combination of forecasts,” *Journal of the Operational Research Society*, vol. 20, no. 4, pp. 451–468, 1969.
- [11] A. Timmermann, “Forecast combinations,” in *Handbook of Economic Forecasting*, vol. 1, G. Elliott, C. W. J. Granger, and A. Timmermann, Eds. Elsevier, 2006, ch. 4, pp. 135–196.
- [12] S. Ben-David, J. Blitzer, K. Crammer, A. Kulesza, F. Pereira, and J. W. Vaughan, “A theory of learning from different domains,” *Machine Learning*, vol. 79, no. 1–2, pp. 151–175, 2010.
- [13] S. L. Braunstein and C. M. Caves, “Statistical distance and the geometry of quantum states,” *Physical Review Letters*, vol. 72, no. 22, pp. 3439–3443, 1994. doi:10.1103/PhysRevLett.72.3439
- [14] C. W. Helstrom, *Quantum Detection and Estimation Theory*. Academic Press, 1976.
- [15] D. C. Brody and L. P. Hughston, “Interest rates and information geometry,” *Proceedings of the Royal Society A*, vol. 457, no. 2010, pp. 1343–1363, 2001. doi:10.1098/rspa.2000.0722
- [16] S. Taylor, “Clustering financial return distributions using the Fisher information metric,” *Entropy*, vol. 21, no. 2, p. 110, 2019. doi:10.3390/e21020110
- [17] B. Horvath, Z. Issa, and A. Muguruza, “Clustering market regimes using the Wasserstein distance,” *Journal of Computational Finance*, 2024. doi:10.21314/jcf.2024.005
- [18] M. Gidea and Y. A. Katz, “Topological data analysis of financial time series: Landscapes of crashes,” *Physica A*, vol. 491, pp. 820–834, 2018. doi:10.1016/j.physa.2017.09.028
- [19] R. S. Sandhu, T. T. Georgiou, E. Reznik, L. Zhu, I. Kolesov, Y. Senbabaoglu, and A. Tannenbaum, “Ricci curvature: An economic indicator for market fragility and systemic risk,” *Science Advances*, vol. 2, no. 6, e1501495, 2016. doi:10.1126/sciadv.1501495
- [20] L. Laloux, P. Cizeau, J.-P. Bouchaud, and M. Potters, “Noise dressing of financial correlation matrices,” *Physical Review Letters*, vol. 83, no. 7, pp. 1467–1470, 1999. doi:10.1103/PhysRevLett.83.1467
- [21] V. Plerou, P. Gopikrishnan, B. Rosenow, L. A. N. Amaral, and H. E. Stanley, “A random matrix approach to cross-correlations in financial data,” *Physical Review E*, vol. 65, 066126, 2002. doi:10.1103/PhysRevE.65.066126
- [22] J. J. Hopfield, “Neural networks and physical systems with emergent collective computational abilities,” *Proc. Natl. Acad. Sci.*, vol. 79, pp. 2554–2558, 1982.
- [23] E. M. Stoudenmire and D. J. Schwab, “Supervised learning with quantum-inspired tensor networks,” in *Advances in Neural Information Processing Systems 29* (NeurIPS 2016), pp. 4799–4807, 2016.
- [24] D. Sornette, “Critical market crashes,” *Physics Reports*, vol. 378, no. 1, pp. 1–98, 2003.

- [25] M. Schuld and N. Killoran, “Quantum machine learning in feature Hilbert spaces,” *Physical Review Letters*, vol. 122, 040504, 2019. doi:10.1103/PhysRevLett.122.040504
- [26] S. Pancharatnam, “Generalized theory of interference, and its applications,” *Proc. Indian Acad. Sci. A*, vol. 44, pp. 247–262, 1956. doi:10.1007/BF03046050
- [27] B. Simon, “Holonomy, the quantum adiabatic theorem, and Berry’s phase,” *Physical Review Letters*, vol. 51, no. 24, pp. 2167–2170, 1983. doi:10.1103/PhysRevLett.51.2167
- [28] A. G. Abanov, L. Candelori, H. C. Steinacker, M. T. Wells, J. R. Busemeyer, V. Kirakosyan, N. Marzari, S. Pinnamaneni, D. Villani, M. Xu, and K. Musaelian, “Quantum geometry of data,” arXiv preprint arXiv:2507.21135, 2025.
- [29] J. Rosaler, L. Candelori, V. Kirakosyan, K. Musaelian, R. Samson, M. T. Wells, D. Mehta, and S. Pasquali, “Supervised similarity for high-yield corporate bonds with quantum cognition machine learning,” arXiv preprint arXiv:2502.01495, 2025.
- [30] R. Samson, A. Banner, L. Candelori, S. Cottrell, T. Di Matteo, P. Duchnowski, V. Kirakosyan, J. Marques, K. Musaelian, S. Pasquali, R. Stever, and D. Villani, “Supervised similarity for firm linkages,” arXiv preprint arXiv:2506.19856, 2025.
- [31] G. Di Caro, V. Kirakosyan, A. G. Abanov, J. R. Busemeyer, L. Candelori, N. Hartmann, E. T. Lam, K. Musaelian, R. Samson, H. Steinacker, D. Villani, M. T. Wells, R. J. Wenstrup, and M. Xu, “Quantum cognition machine learning for forecasting chromosomal instability,” arXiv preprint arXiv:2506.03199, 2025.
- [32] R. P. Adams and D. J. C. MacKay, “Bayesian online changepoint detection,” arXiv preprint arXiv:0710.3742, 2007.
- [33] Z. Issa and B. Horvath, “Non-parametric online market regime detection and regime clustering for multidimensional and path-dependent data structures,” arXiv preprint arXiv:2306.15835, 2023.
- [34] M. M. Bronstein, J. Bruna, T. Cohen, and P. Veličković, “Geometric deep learning: Grids, groups, graphs, geodesics, and gauges,” arXiv preprint arXiv:2104.13478, 2021.
- [35] K. Musaelian, A. Abanov, J. Berger, L. Candelori, V. Kirakosyan, R. Samson, J. Smith, and D. Villani, “Quantum cognition machine learning: AI needs quantum,” Qognitive, Inc., technical white paper (not peer-reviewed), 2024.
- [36] R. Samson, J. Berger, L. Candelori, V. Kirakosyan, K. Musaelian, and D. Villani, “Quantum cognition machine learning: Financial forecasting,” Qognitive, Inc., technical white paper (not peer-reviewed), 2024.
- [37] E. S. Page, “Continuous inspection schemes,” *Biometrika*, vol. 41, no. 1/2, pp. 100–115, 1954. doi:10.2307/2333009

A Proofs of Formal Results

Proof of Theorem 1. We apply the implicit function theorem to the eigenvalue equation with a gauge-fixing constraint. Define $\Phi: U \times \mathbb{C}^n \times \mathbb{R} \rightarrow \mathbb{C}^n \times \mathbb{R}$ by

$$\Phi(x, \psi, E) = (H(x)\psi - E\psi, \langle \psi_0(x_0) | \psi \rangle - 1),$$

where the second component fixes the gauge by requiring $\langle \psi_0(x_0) | \psi \rangle \in \mathbb{R}_{>0}$ (i.e., real positive overlap with the reference state at a base point x_0). At a solution (x_0, ψ_0, E_0) the Jacobian with respect to (ψ, E) is

$$D_{(\psi, E)}\Phi|_{(x_0, \psi_0, E_0)} = \begin{pmatrix} H(x_0) - E_0 I & -\psi_0 \\ \langle \psi_0 | & 0 \end{pmatrix}.$$

Decompose $\mathbb{C}^n = \text{span}\{\psi_0\} \oplus \psi_0^\perp$. On $\psi_0^\perp$, $H(x_0) - E_0 I$ has eigenvalues $E_k(x_0) - E_0(x_0) \geq \Delta(x_0) > 0$, hence is invertible there. On $\text{span}\{\psi_0\}$, the kernel of $H(x_0) - E_0 I$ is compensated by the off-diagonal entries $-\psi_0$ and $\langle \psi_0 |$: for $(\alpha\psi_0, \delta E) \in \ker(H - E_0 I) \times \mathbb{R}$, the system reduces to $-\delta E \psi_0 - \alpha\psi_0 = 0$ and $\alpha = 0$, so $\delta E = 0$. Thus $D_{(\psi, E)}\Phi$ is invertible.

The implicit function theorem [8] then gives C^∞ maps $x \mapsto (\psi(x), E_0(x))$ on U . Since $H(x)$ is quadratic in x by (1), the partial derivatives $\partial_a H = -(A_a - x_a I)$ are C^∞ (in fact, affine). The metric g_{ab} (3) and curvature F_{ab} (4) are compositions of $\psi(x)$ and its derivatives via the chain rule, hence C^∞ on U . $\square$

Theorem 3 (Chern number quantization [28]). *Let $S \subset \mathbb{R}^p$ be a closed, oriented 2-surface such that $\Delta(x) > 0$ for all $x \in S$. Then*

$$C_1 = \frac{1}{2\pi} \int_S F_{ab} dx^a \wedge dx^b \in \mathbb{Z}. \quad (8)$$

Proof. By Theorem 1, the ground state $|\psi(x)\rangle$ is C^∞ on S . Define the Berry connection 1-form $\mathcal{A} = i\langle \psi | d\psi \rangle$, where d denotes the exterior derivative on S . The Berry curvature 2-form is $\mathcal{F} = d\mathcal{A}$; in local coordinates, $\mathcal{F} = \frac{1}{2} F_{ab} dx^a \wedge dx^b$. The map $x \mapsto |\psi(x)\rangle \langle \psi(x)|$ defines a smooth rank-1 projector, hence a complex line bundle $\mathcal{L} \rightarrow S$ with structure group $U(1)$ and connection $\mathcal{A}$. By the Chern–Weil theorem [5, 9], the first Chern number $C_1 = \frac{1}{2\pi} \int_S \mathcal{F}$ is an integer, since it equals the degree of the classifying map $S \rightarrow \mathbb{CP}^{n-1}$. $\square$

Caveat. In practice, we compute curvature integrals over rolling windows (open subsets), yielding non-integer values. Topological protection does not strictly apply.

Theorem 4 (Curvature–gap bound). *For all $x \in U$ and indices a, b ,*

$$|F_{ab}(x)| \leq \frac{2 \|\partial_a H(x)\|_{\text{op}} \|\partial_b H(x)\|_{\text{op}}}{\Delta(x)^2}, \quad (9)$$

where $\|\cdot\|_{\text{op}}$ denotes the operator norm. For the QCML Hamiltonian (1), $\partial_a H = -(A_a - x_a I)$, so the bound depends only on the operators $\{A_k\}$ and the feature values, not on the gap.

Proof. First-order perturbation theory gives the exact representation

$$F_{ab} = -2 \text{Im} \sum_{n \geq 1} \frac{\langle \psi_0 | \partial_a H | \psi_n \rangle \langle \psi_n | \partial_b H | \psi_0 \rangle}{(E_n - E_0)^2}, \quad (10)$$

where $\{|\psi_n\rangle\}_{n\geq 0}$ is a complete eigenbasis of $H(x)$. Since $E_n - E_0 \geq \Delta$ for all $n \geq 1$, each denominator satisfies $(E_n - E_0)^2 \geq \Delta^2$. Hence

$$|F_{ab}| \leq \frac{2}{\Delta^2} \sum_{n\geq 1} |\langle \psi_0 | \partial_a H | \psi_n \rangle| |\langle \psi_n | \partial_b H | \psi_0 \rangle|.$$

By the Cauchy–Schwarz inequality on the sum,

$$\sum_{n\geq 1} |\langle \psi_0 | \partial_a H | \psi_n \rangle| |\langle \psi_n | \partial_b H | \psi_0 \rangle| \leq \left(\sum_{n\geq 1} |\langle \psi_0 | \partial_a H | \psi_n \rangle|^2 \right)^{1/2} \left(\sum_{n\geq 1} |\langle \psi_n | \partial_b H | \psi_0 \rangle|^2 \right)^{1/2}.$$

By the completeness relation $\sum_{n\geq 0} |\psi_n\rangle\langle\psi_n| = I$, each factor is bounded by the operator norm: $\sum_{n\geq 1} |\langle \psi_0 | \partial_a H | \psi_n \rangle|^2 \leq \|\partial_a H | \psi_0\rangle\|^2 \leq \|\partial_a H\|_{\text{op}}^2$. Combining yields (9). $\square$

Proposition 5 (QFI volume). *Let g be a $p \times p$ positive semi-definite matrix of rank $r \leq p$, with positive eigenvalues $\lambda_1 \geq \dots \geq \lambda_r > 0$. Define the pseudo-determinant*

$$\det^+(g) = \prod_{i=1}^r \lambda_i. \quad (11)$$

Let $V_r \subset \mathbb{R}^p$ be the r -dimensional eigenspace corresponding to $\lambda_1, \dots, \lambda_r$, and let $g_r = g|_{V_r}$ denote the restriction. Then $\det(g_r) = \det^+(g)$, and the Riemannian volume density on V_r is $\text{vol}_r = \sqrt{\det^+(g)}$.

Proof. By the spectral theorem, g admits an orthonormal eigenbasis $\{e_1, \dots, e_p\}$ with $g e_i = \lambda_i e_i$, where $\lambda_i > 0$ for $i \leq r$ and $\lambda_i = 0$ for $i > r$. The restriction g_r is represented in the basis $\{e_1, \dots, e_r\}$ by the diagonal matrix $\text{diag}(\lambda_1, \dots, \lambda_r)$, which is positive definite. Therefore $\det(g_r) = \prod_{i=1}^r \lambda_i = \det^+(g)$. The Riemannian volume density on the r -dimensional submanifold equipped with metric g_r is $\text{vol}_r = \sqrt{\det(g_r)} = \sqrt{\det^+(g)}$, the standard formula for the volume element of a Riemannian metric [7]. $\square$

B Granger Causality

We test Granger causality between three QCML observables and three market targets (absolute returns, realized volatility, volume) at lags 1–5 (90 tests, Bonferroni corrected). QFI Determinant Granger-causes absolute returns at lag 1 ($F = 9.5$, $p = 0.0004$, Bonferroni-significant). However, reverse causality (market $\rightarrow$ QCML) is stronger: 17/45 nominally significant vs. 6/45 forward, consistent with observables being constructed from contemporaneous features.

C Default Hyperparameters

D Numerical Stability Ablation

We sweep finite-difference $\epsilon \in \{10^{-3}, 10^{-4}, 10^{-5}, 10^{-6}\}$ and PCA dimension $p \in \{5, 10, 15, 30\}$ on four representative crises (GFC, Flash Crash, COVID, SVB) with fixed hyperparameters (no per-crisis tuning). Results appear in Table 17. Berry curvature is extremely stable across ϵ (all median $d \approx 0.15$), while QFI determinant shows moderate sensitivity ($d = 0.52$ at $\epsilon = 10^{-3}$ vs. $d = 0.36$ at 10^{-5}), saturating below 10^{-5} . Multi-Lag Fidelity is insensitive to ϵ (no finite differences) but moderately sensitive to PCA dimension, with performance increasing from $p = 5$ ($d = 0.29$) to $p = 10$ ($d = 0.44$) and stabilizing thereafter.

Table 16: Default hyperparameters.

Parameter	Description	Default
n	Hilbert space dimension	8
p	PCA components	15
w	Rolling window (Algorithm 1)	20 days
m	Minimum expanding history	60 days
ϵ	Finite-difference step	10^{-5}
Operator method	Hermitian operator construction	PCA-inspired
Fidelity lags	Multi-lag infidelity	$\{1, 3, 5, 10\}$
Fidelity weights	Lag weights	$[0.4, 0.3, 0.2, 0.1]$

Table 17: Numerical stability: median Cohen’s d across 4 crises. Multi-Lag Fidelity does not use ϵ (no finite differences).

Config	Berry	QFI Det.	Multi-Lag
$\epsilon = 10^{-3}$	0.15	0.52	—
$\epsilon = 10^{-4}$	0.15	0.60	—
$\epsilon = 10^{-5}$ (default)	0.15	0.36	—
$\epsilon = 10^{-6}$	0.15	0.36	—
$p = 5$	0.09	0.34	0.29
$p = 10$	0.20	0.41	0.44
$p = 15$ (default)	0.15	0.36	0.45
$p = 30$	0.22	0.54	0.45

E Window-Size Sensitivity

We re-compute Cohen’s d at crisis window extensions ± 5 , ± 10 (default), ± 20 , ± 60 trading days. Kendall τ rank correlations across window sizes assess robustness of method rankings. Results appear in Table 18. Rankings are stable for nearby windows ($\tau = 0.87$ for ± 5 vs. ± 10 and ± 5 vs. ± 20) but degrade at ± 60 days ($\tau = 0.07$ – 0.33), where diluted crisis signal reduces separation. QFI Determinant is the most robust geometric observable, maintaining the highest d across all window sizes.

Table 18: Median Cohen’s d by crisis window size (14 crises).

Method	$\pm 5d$	$\pm 10d$	$\pm 20d$	$\pm 60d$
Berry Curv. Incr.	0.25	0.29	0.27	0.24
QFI Determinant	0.63	0.36	0.64	0.43
Multi-Lag Fidelity	0.40	0.27	0.38	0.26
Random Forest	0.21	0.21	0.28	0.34

F Hyperparameter Sensitivity: Fixed vs. Tuned

Table 19 compares three hyperparameter regimes: (a) fixed defaults with all `pca_inspired` operators (the previous default), (b) fixed defaults with per-method operator selection (Berry and QFI use

`random`; MLF uses `pauli`—these are the main results in Table 4), and (c) Optuna HPO (100 trials, pre-2020 training crises, post-2020 validation). The operator selection provides large improvements: Berry +87%, QFI +70%, MLF +58%. Naïve HPO showed severe overfitting (train d exceeding validation by 0.35–0.75); regularized HPO with a consistency penalty reduces this gap to ≤ 0.20 , with QFI achieving val $d = 0.60 > \text{train } d = 0.50$ (column c^*).

Table 19: Hyperparameter sensitivity: median Cohen’s d across 14 crises (columns a–b) and train/val split (column c).

Method	(a) All PCA-Insp.	(b) Best Operator	(c) HPO Val.
Berry Curvature Increment	0.29	0.55	0.49
QFI Determinant	0.36	0.61	0.60
Multi-Lag Fidelity	0.27	0.42	0.33

(a,b) Fixed $n = 8$, $p = 15$, $w = 20$. (b) Berry/QFI: `random`; MLF: `pauli` ($\text{exp}=0.0$). (c^*) Regularized Optuna TPE, 100 trials; consistency penalty $\text{mean}_d - 0.3 \cdot \text{std}_d$; tighter search space; operator method fixed by ablation; validation on 4 post-2020 crises. Train d : Berry 0.68, QFI 0.50, MLF 0.53. Train–val gap ≤ 0.20 (vs. 0.35–0.75 without regularization).

G Online Detection and PnL Backtest

To evaluate practical deployability, we implement a strictly causal online pipeline that computes geometric features and $P(\text{crisis})$ at each timestep using only data available up to that point. The pipeline comprises: (i) an *OnlineGeometricFeatureComputer* that periodically refits scaler/PCA/operators on a rolling 500-day window and outputs 10 features (5 base geometric observables plus 5 EMA velocity features); (ii) three detectors—multivariate Bayesian filtering, periodic HMM refit, and expanding-percentile RMS scoring—combined via a weighted ensemble (weights 0.4/0.4/0.2).

Table 20: Online detection performance (4,934 timesteps, 2005–2024). Detection rate and delay measured at 1 false alarm per year.

Detector	AUC-ROC	AUC-PR	Det@1/yr	Delay
Online CUSUM	0.619	0.330	67%	39d
Online Vol-Z	0.576	0.366	67%	20d
Online Logistic (supervised)	0.528	0.261	33%	82d
Online Ensemble	0.502	0.221	17%	20d
Online Percentile	0.502	0.218	0%	—
Online Bayesian	0.500	0.221	0%	—

The naïve online results are sobering: the geometric ensemble achieves only $\text{AUC-ROC} = 0.502$, barely above chance. Diagnostic analysis reveals that geometric features have *lower* mean values during crises—the wrong direction for anomaly scoring. The effect sizes are meaningful (QFI: $d = 1.79$; spectral gap: $d = 1.48$) but inverted.

Signal inversion fix. Once the directional mismatch is identified, the fix is straightforward: negate features whose crisis-period mean is lower than the normal-period mean. With this correction, the expanding-percentile detector achieves $\text{AUC-ROC} = 0.84$, a 68% improvement over the

naïve pipeline and 36% over classical CUSUM (0.62). An unsupervised variant (negating all features without labels) achieves the same AUC-ROC = 0.84, confirming the correction is robust. The directional mismatch arises because crises correspond to manifold *contraction*—lower curvature and fidelity—rather than the expansion assumed by standard anomaly detectors.

PnL Backtest. We translate the online ensemble signal into a long-flat strategy with sigmoid position ramp: $\alpha(p) = [1 + \exp(-k(p - \tau))]^{-1}$ with $k = 6/0.3 = 20$ and threshold $\tau = 0.5$. The strategy targets 10% annualized volatility, applies 0.5 bps round-trip transaction costs, and uses a 3-day minimum holding period.

Table 21: Backtest results (2005–2024). IS/OOS split at 2020-01-01.

Strategy	Sharpe	IS Sharpe	OOS Sharpe	MaxDD
ConstantVol SPY	0.63	0.57	0.79	27.0%
Buy & Hold SPY	0.51	0.45	0.68	56.5%
Geometric LongFlat	0.47	0.36	0.79	19.0%
Geometric LongShort	0.40	0.27	0.81	18.2%

The geometric strategy achieves comparable OOS Sharpe (0.79 vs. 0.79 for ConstantVol) while reducing maximum drawdown from 27.0% to 19.0% (−30% relative) and from 56.5% to 19.0% versus buy-and-hold (−66% relative). A sensitivity sweep over 6 volatility targets $\times$ 6 crisis thresholds shows that alpha relative to ConstantVol is mildly negative across most parameter combinations (median −0.10) but approaches zero at higher thresholds (≥ 0.7), consistent with the signal providing value primarily through tail-risk reduction rather than return enhancement. The break-even transaction cost is approximately 1 bps.

H Normalization Ablation

Table 22 compares four normalization modes for the PCA-projected features before Hilbert space embedding: *sphere* (L^2 -normalization to the unit sphere), *none* (raw PCA components), *soft* ($x/(\|x\| + \tilde{m})$ where $\tilde{m}$ is the training-set median norm), and *clip* (component-wise clipping to $[-3\sigma, 3\sigma]$ followed by L^2 -normalization). Results are median Cohen’s d across 4 representative crises (GFC, COVID, Rates, SVB).

Table 22: Normalization ablation: median Cohen’s d across 4 crises. Bold = best mode per detector.

Mode	Berry	QFI Det.	Multi-Lag
Sphere	0.54	0.44	0.63
None	0.46	1.45	0.53
Soft	0.66	1.18	0.57
Clip	0.46	1.40	0.58

Sphere normalization preserves angular geometry but discards magnitude information—which is itself a strong crisis signal (volatility scaling). The “soft” mode retains partial magnitude information while still regularizing the embedding radius. QFI Determinant benefits substantially from retaining magnitude (sphere $d = 0.44$ vs. none $d = 1.45$, a $3.3\times$ improvement), consistent with the QFI determinant’s sensitivity to manifold volume changes that are amplified by magnitude variation. Berry curvature increment performs best under soft normalization ($d = 0.66$), while Multi-Lag

Fidelity is most robust under sphere normalization ($d = 0.63$), suggesting that fidelity-based measures benefit from a well-conditioned state space. Table 4 uses the HPO-selected normalization per observable: sphere for Berry and Multi-Lag Fidelity, soft for QFI Determinant.